Central-depth stability and optimal central realizations of three-Engel Lie algebras

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Abstract

Let k have characteristic two and more than two elements. For ordinary three-Engel Lie algebras with second derived ideal in the sth upper center, let a_s be the least eventual starting rank of the class law $d+2$. We prove that every a_s exists and, for $s \geq 3$, $s + 2^{\lfloor \log_2(s-1) \rfloor} \leq a_s \leq 7s + 2^{\lceil \log_2 s \rceil} - 9 \leq 9s - 11$. A fixed-anchor Frobenius-jet collecting argument gives the upper bound; cyclic-support Zassenhaus currents give the lower bound and an exact depth filtration. Thus a_s and the least complete-bottom class-raising lift delay have linear order. We retain the sharp endpoints $a_5 = 9$, $a_6 = 10$ and construct a determinant-twisted symmetric-cube quotient at every depth. We also determine the least center realizing the free metabelian quotient and natural graph quotients. The arbitrary-depth proof is independent of the finite starting systems used for the exact low-depth endpoints.

Keywords. Three-Engel Lie algebra; nilpotency class; central depth; central realization; Koszul module; cubic representation.

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1 Introduction

Let k have characteristic two and $|k| > 2$. We study ordinary Lie algebras satisfying $[y, x, x, x] = 0$. The central question is how the depth of the second derived ideal controls nilpotency class as the number of generators grows. Write

$$M_0(H) = H'', \qquad M_{i+1}(H) = [M_i(H), H], \qquad D_{d,s} = \mathcal{F}_d/M_s(\mathcal{F}_d),$$

where $\mathcal{F}_d$ is relatively free on d ordinary generators. Thus $D_{d,s}$ is universal under $H'' \subseteq Z_s(H)$.

1.1 Uniform stability at every depth

For each $s \geq 1$, let a_s be the least rank from which $\operatorname{cl} D_{d,s} = d + 2$ holds for all larger ranks. We prove that every a_s exists and that, for $s \geq 3$,

$$s + 2^{\lfloor \log_2(s-1) \rfloor} \leq a_s \leq 7s + 2^{\lceil \log_2 s \rceil} - 9 \leq 9s - 11.$$

In particular $a_s = \Theta(s)$. The upper and lower bounds come from different structural mechanisms. The upper bound uses fixed-anchor partial Engel identities and a Frobenius-jet collecting argument. The lower bound uses a cyclic-support current over a classical derived Zassenhaus algebra, with an exact two-parameter depth law. Neither bound is inferred from a table of finite ranks.

The collecting argument works in an auxiliary Lie algebra where the anonymous singleton letter has unrestricted adjoint action. High jets generate an ideal centralizing the second derived ideal. A simultaneous square relation bounds the anchor degree of its bracket square. Lazard elimination then gives a uniform cone of vanishing components. Keeping one distinguished singleton produces a surjection onto all three mixed critical profiles, and induction on central depth gives the stated stable rank bound. This proof does not require the finite starting systems used for the sharper low-depth endpoints.

The current family has exact ordinary rank $m + 1$, one-dimensional center, and

$$\operatorname{cl} J_{q,m} = m + \left\lfloor \frac{m-1}{q-1} \right\rfloor + 2, \qquad M_u(J_{q,m}) = \gamma_{q+u+3}(J_{q,m}), \quad q = 2^r \geq 2.$$

Its central truncations supply every-depth lower witnesses. Affine frame automorphisms show that the witnesses are critical for their class: every proper subalgebra and proper quotient has smaller class. For the complete bottom $E_d = D_{d,1}$, the least number $R(d)$ of rank-preserving ordinary central lifts needed to exceed class $d + 2$ also has linear order. More explicitly, for $d \geq 7$,

$$\left\lfloor \frac{d+11}{9} \right\rfloor \leq R(d) \leq d - q_d < \frac{3d}{4},$$

where q_d is the largest power of two at most $d/2$.

1.2 Exact boundaries and critical representations

The general bound complements sharp fixed-depth information. For central second derived ideal, the exact class maximum is $\max(5, d + 2)$. The ceiling $d + 2$ persists at depths two, three and four from ranks four, five and six, respectively. The least starts at depths five and six are $a_5 = 9$ and $a_6 = 10$. The corresponding complete-bottom delays include $R(9) = 5$ and $R(10) = 6$. Their sharper upper ranges use balanced-module collection, rank-uniform propagation, and explicit finite starting systems.

At the fifth-depth boundary the rank-seven critical layer is $\det(U) \otimes \mathrm{Lie}^3(U)$, while rank eight gives $\det(U) \otimes \mathrm{Sym}^3(U)$ with its nonsplit Frobenius structure. At every depth $s \geq 3$, set $d = s + 2^{\lfloor \log_2(s-1) \rfloor} - 1$. We construct an algebraic $\mathrm{GL}(U)$-equivariant quotient

$$(D_{d,s})_{d+3} \twoheadrightarrow \det(U) \otimes \mathrm{Sym}^3(U).$$

This is an actual Lie quotient realization; equality with the whole free critical layer is asserted only at the previously determined rank-eight boundary.

1.3 Optimal centers and context

For the free metabelian three-Engel algebra H_d, $d \geq 3$, we prove

$$\min_{E/Z(E) \simeq H_d} \dim_k Z(E) = \left\lceil \frac{d+1}{2} \right\rceil.$$

The classical center-to-Hom injection gives the lower bound. Pair detectors and a top symmetric cocycle attain it in an ordinary three-Engel algebra with $Z(E) = E''$. For natural class-four graph quotients Q_G, local Jacobi restrictions give the exact minimum $\max_{\varnothing \neq W \subseteq V(G)} \lceil |E(G[W])|/|W| \rceil$, attained by a class-five realization.

The ordinary Engel theory and generator-dependent class bounds have the antecedents [7, 8]. The commutator and balanced-module framework is classical [2, 1, 3]; so are Lazard elimination [9], powers of derivations, the Zassenhaus input [10], and the cubic representations. Here these inputs give the fixed-anchor jet cone, all mixed critical weights, exact current filtrations, and the stated stable ranks and central realizations.

Quadratic orbit maps and their additional reconstruction conditions are treated in [12]; metabelian extremal dimensions and type growth are treated in [13]. These are comparison references, developed from the common research collection [14]. The ordinary theorems here require no quadratic $R = 0$ realization and no new result from those companion papers. The graph realization problem is likewise distinct from counting ideals in a graph source.

Sections 2 and 3 give the commutator and central-cover foundations. Section 4 states the uniform depth laws and exact endpoints; Section 5 treats prescribed central realizations. The appendices separate the Frobenius-jet and current proofs from the finite endpoint systems. All generators are ordinary and dimensions are over k; brackets are alternating and left-normed. The optimal growth of the unrestricted class maximum $e(d)$ remains a separate problem.

2 The ordinary metabelian model

Work over a characteristic-two field k with $|k| > 2$.

2.1 A multigraded commutator module

For $d \geq 2$, put

$$R_d = k[t_1, \ldots, t_d]/(t_1^2, \ldots, t_d^2), \qquad \mathfrak{m} = (t_1, \ldots, t_d).$$

Let W_d be the R_d-module generated by v_{ij}, $i < j$, subject to

$$t_i v_{jk} + t_j v_{ik} + t_k v_{ij} = 0 \qquad (i < j < k). \tag{1}$$

We set $v_{ji} = v_{ij}$ and $v_{ii} = 0$. This is the classical free metabelian commutator-module presentation, base-changed to R_d; compare [5, Theorem 6.2] and [1, Sections 2–3].

The ordinary algebra $H_d = U \oplus W_d$, with $[x_i, x_j] = v_{ij}$, $[x_i, w] = t_i w$ and $[W_d, W_d] = 0$, is the free metabelian three-Engel algebra. In a metabelian target the derived actions commute.

Partial linearization of $[y,x,x,x]=0$ over $|k|>2$ forces $t_x^2(H')=0$, and Jacobi gives the triangles. Conversely these conditions imply three-Engel. Its lower-central terms are

$$\gamma_j(H_d)=\mathfrak{m}^{j-2}W_d \qquad (j\geq 2). \tag{2}$$

Grade R_d by coefficient degree, and put the generators v_{ij} in coefficient degree zero. We also use the finer grading $\deg t_i=e_i$, $\deg v_{ij}=e_i+e_j$.

Proposition 2.1. *The coefficient-degree Hilbert polynomial of W_d is*

$$H_d(T)=\sum_{k=2}^{d}\binom{d}{k}\big((k-1)T^{k-2}+kT^{k-1}\big)+\binom{d}{2}T^2. \tag{3}$$

Consequently

$$\dim_k W_d=(d-1)2^d+1-d+\binom{d}{2}.$$

The ordinary algebra H_d has class $\max(4,d+1)$.

Proof. A raw multihomogeneous monomial is $t_S v_{ij}$, where S is square-free. Its multidegree has entries zero, one, or two, and at most two entries are doubled. Fix its support of size $k\geq 2$.

If no index is doubled, the raw coordinates are the edges of the complete graph on the support. Relations (1) are its triangle sums. The quotient edge space has dimension $k-1$, in coefficient degree $k-2$.

If exactly one index j is doubled, only edges incident with j occur. Triangle relations through j identify every pair of these edges, leaving dimension one. There are k choices of j, in coefficient degree $k-1$.

If exactly two indices i,j are doubled, there is one raw coordinate. A third index in the support kills it by a triangle relation, since the other two terms contain a squared variable. The coordinate survives exactly when the support is $\{i,j\}$, giving the final term of (3). These cases exhaust the multigrading.

The maximum coefficient degree is two for $d=2,3$, and $d-1$ for $d\geq 4$. Equation (2) gives the class. □

This identifies the top layers used to study ordinary central extensions.

3 Top central covers and a sharp class boundary

We study the first nonmetabelian boundary in ordinary three-Engel theory. Put $U=k^d$ and use the classical commutator module W_d of Section 2. The ordinary Lie algebra

$$H_d=U\oplus W_d, \qquad [x_i,x_j]=v_{ij}, \quad [x_i,w]=t_i w, \quad [W_d,W_d]=0 \tag{4}$$

is the universal metabelian 3-Engel algebra on d ordinary generators. Give x_i Lie weight one and v_{ij} weight two. For $d\geq 4$, its top weight is $d+1$.

Theorem 3.1 (The full top central cover)**.** *Let $d\geq 4$, and give a central coefficient space Z weight $d+2$. There is a natural isomorphism*

$$H^2_{\mathrm{CE}}(H_d,Z)_{d+2}\simeq \mathrm{Hom}_k\big(\det(U)\otimes \mathrm{Sym}^2 U,Z\big), \tag{5}$$

where the subscript denotes total input weight. Every class in this homogeneous space defines an ordinary 3-Engel central extension. If $\dim_k Z=1$, every nonzero class gives an extension of exact ordinary generator rank d, class $d+2$, and second derived ideal equal to Z.

The top commutator space is naturally $\det(U)\otimes U$. The CE equations determine a cocycle from its pairing of that space with U, and the pairing is exactly a symmetric bilinear form with the displayed determinant twist. The symmetry is bilinear symmetry in characteristic two, so its diagonal coefficients remain independent.

Let $e_{\mathrm{cbm}}(d)$ be the largest class of a finite-dimensional ordinary 3-Engel k-Lie algebra of generator rank at most d satisfying $L'' \subseteq Z(L)$. Thus $L/Z(L)$ is metabelian.

Theorem 3.2 (A sharp centre-by-metabelian class law)**.** *For every $d \geq 2$,*

$$e_{\mathrm{cbm}}(d) = \max(5, d+2). \tag{6}$$

The maximum is also attained at exact ordinary generator rank d.

The upper bound is the classical metabelian ceiling $\max(4, d+1)$, obtained from W_d, followed by one central step. The top covers attain it for $d \geq 4$. A direct two-generator calculation supplies the low-rank endpoints and proves $e(2) = 5$ (Lemma A.2). Consequently any superlinear mechanism for $e(d)$ must have noncentral second derived ideal.

The commutator module and the CE/Hopf central-cover framework are classical; the free centre-by-metabelian setting is developed in [1]. Here the square-zero module relations determine the full top quotient and make its cocycles automatically 3-Engel. Appendix A gives the quotient calculation, the ordinary metabelian ceiling, and the attaining models.

4 Uniform central depth and exact endpoints

Let $\mathcal{F}_d$ be the relatively free ordinary three-Engel algebra. Set $M_0 = \mathcal{F}_d''$, $M_{j+1} = [M_j, \mathcal{F}_d]$, $D_{d,s} = \mathcal{F}_d/M_s$, and $E_d = D_{d,1}$. The condition $M_s(H) = 0$ is equivalent to $H'' \subseteq Z_s(H)$. Let a_s denote the least eventual starting rank of $\operatorname{cl} D_{d,s} = d+2$.

Theorem 4.1 (Every-depth stability and linear growth)**.** *Every a_s, $s \geq 1$, exists. Put $Q(s) = 2^{\lceil \log_2 s \rceil}$. For $s \geq 2$,*

$$\operatorname{cl} D_{d,s} = d+2 \quad \text{if } d \geq 7s + Q(s) - 9.$$

For $s \geq 3$,

$$s + 2^{\lfloor \log_2(s-1) \rfloor} \leq a_s \leq 7s + Q(s) - 9 \leq 9s - 11.$$

Thus $a_s = \Theta(s)$. More generally, a stable $(s-1)$-depth ceiling from rank a implies an s-depth ceiling from rank

$$\max\{a+1,\ 7s + Q(s) - 9\}. \tag{7}$$

Every ceiling is attained at exact ordinary generator rank d.

Appendix B proves the upper bound by a fixed-anchor Frobenius-jet argument. Its auxiliary singleton letter is not required to be Engel. Keeping one distinguished singleton gives a surjection onto each mixed critical weight. The exact current construction in Appendix C proves the lower bound: for every $s \geq 3$ there is a one-dimensional-center witness of rank $s + 2^{\lfloor \log_2(s-1) \rfloor} - 1$, class three above that rank, and exact depth s. The two arguments form an all-depth proof; the finite systems below are used for sharper exact endpoints.

4.1 Sharp low-depth ranges

Theorem 4.2 (Depths two, three, and four)**.** *For an at-most-d-generated ordinary three-Engel algebra,*

$$H'' \subseteq Z_s(H) \Longrightarrow \operatorname{cl} H \leq d+2 \quad \begin{cases} s = 2, & d \geq 4, \\ s = 3, & d \geq 5, \\ s = 4, & d \geq 6. \end{cases}$$

Each bound is attained at every exact rank in its stated range.

The balanced symmetric-square and peeling arguments are in Appendix E. At depth three, two polarized Engel instances and thirteen depth relations give an explicit collecting identity. The rank-six fourth-depth starting step has three complete finite systems; symbolic suspension propagates it to arbitrary rank.

Proposition 4.3 (Boundary ranks)**.** *The complete relatively free objects have classes*

	$D_{d,1}$	$D_{d,2}$	$D_{d,3}$	$D_{d,4}$
$d=4$	6	6	7	*not determined here*
$d=5$	7	7	7	8

At rank five there are central epimorphisms

$$A_{178} \twoheadrightarrow A_{167} \twoheadrightarrow A_{139} \twoheadrightarrow A_{118}, \tag{8}$$

of classes $8,7,7,7$*, with kernels of dimensions* $11,28,21$*. At rank four there is a* $42 \to 41 \to 40$ *chain of classes* $7,6,6$*. All objects have the stated exact ordinary generator rank.*

Theorem 4.4 (Exact fifth- and sixth-depth starts)**.** *The stable class law* $\operatorname{cl} D_{d,s} = d+2$ *holds at depth five for* $d \geq 9$ *and at depth six for* $d \geq 10$*. Both starts are least possible:*

$$a_5 = 9, \qquad a_6 = 10.$$

In particular $\operatorname{cl} D_{7,5} = 10$*,* $\operatorname{cl} D_{8,5} = 11$*, and* $\operatorname{cl} D_{9,6} = 12$*.*

The sharp upper ranges follow from the exact starts and source-level propagation in Appendix F. The currents $J_{4,7}$ and $J_{4,8}$ give the rank-eight and rank-nine lower witnesses by Theorem C.2; the latter has dimension 597, class 12, and depth six. The central-step upper bound gives the displayed equality for $D_{9,6}$. These exact endpoint inputs have a separate role from the finite-start-free bound of Theorem 4.1.

4.2 Cubic critical quotients

Theorem 4.5 (The fifth-depth boundary representations)**.** *For* $T_d = (D_{d,5})_{d+3}$ *and* $U = k^d$*,*

$$T_7 \simeq \det(U) \otimes \operatorname{Lie}^3(U), \quad \dim T_7 = 112, \qquad T_8 \simeq \det(U) \otimes \operatorname{Sym}^3(U), \quad \dim T_8 = 120.$$

The rank-eight module has a nonsplit Frobenius submodule of dimension 64 *and exterior quotient of dimension* 56*.*

Theorem 4.6 (A symmetric-cube quotient at every depth)**.** *For* $s \geq 3$*, set* $q = 2^{\lfloor \log_2(s-1) \rfloor}$*,* $d = s + q - 1$*, and* $U = k^d$*. There is an algebraic* $\mathrm{GL}(U)$*-equivariant quotient*

$$(D_{d,s})_{d+3} \twoheadrightarrow \det(U) \otimes \operatorname{Sym}^3(U).$$

It is the full top layer of a d*-generated ordinary three-Engel realization of class* $d+3$ *and exact depth* s*. The quotient has the nonsplit sequence*

$$0 \longrightarrow \det(U) \otimes U^{[1]} \otimes U \longrightarrow \det(U) \otimes \operatorname{Sym}^3 U \longrightarrow \det(U) \otimes \Lambda^3 U \longrightarrow 0.$$

The proof in Appendix D uses the current's affine frame automorphisms and actual root coefficients of a Lie word. The theorem asserts a quotient, not an identification of the whole free component. Finite fields use the algebraic action from formal Engel coefficients. The one-dimensional-center current and the covariant realization with the full cubic top are different algebras.

4.3 The first class-raising central lift

Define $R(d)$ as the least number of successive central epimorphisms over the complete bottom E_d needed to reach class greater than $d+2$, with every algebra generated by at most d ordinary elements.

Theorem 4.7 (Linear delay over the complete bottom)**.** *For $d \geq 7$, let q_d be the largest power of two at most $d/2$. Then*

$$\left\lfloor \frac{d+11}{9} \right\rfloor \leq R(d) \leq d - q_d < \frac{3d}{4}.$$

Thus $R(d) = \Theta(d)$. The upper bound holds already for $d \geq 4$.

Proof. After l lifts, $M_{l+1} = 0$. Put $s = \lfloor (d+11)/9 \rfloor \geq 2$. Theorem 4.1 excludes class growth through $s-1$ lifts. The currents and marked fiber product in Corollary C.7 give the upper bound over the complete bottom. □

Corollary 4.8 (Exact endpoint delays)**.** *One has $R(5) = 3$, $R(7) = R(8) = 4$, $R(9) = 5$, and $R(10) = 6$.*

Proof. The fixed-depth ceilings give the respective lower bounds. The boundary models give attainment at ranks five, seven, and eight. The currents $J_{4,8}$ and $J_{4,9}$ have rank nine and ten, class 12 and 13, and exact depth six and seven. Their marked fiber products with E_d give the final two attaining towers. □

5 Optimal centers in ordinary central realizations

Central realization asks for an algebra whose quotient by its entire center is a prescribed object. For the universal metabelian algebra H_d in (4), the smallest possible center has a sharp formula. The field in the ordinary statements below may be any characteristic-two field with more than two elements.

Theorem 5.1 (The least center in a central realization)**.** *For $d \geq 3$, put $h_d = (d-1)2^d + 1 + \binom{d}{2}$. Then*

$$\min_{E/Z(E) \simeq H_d} \dim_k Z(E) = \left\lceil \frac{d+1}{2} \right\rceil.$$

The minimum is over all finite-dimensional ordinary Lie realizations. It is attained by a d-generated three-Engel algebra of dimension $h_d + \lceil (d+1)/2 \rceil$, class $d+2$, and $Z(E) = E''$. For the complete object $E_d = D_{d,1}$,

$$Z(E_d) = E_d'', \qquad E_d/Z(E_d) \simeq H_d.$$

The lower bound is the classical injection $Z(H_d) \hookrightarrow \mathrm{Hom}_k(H_d/H_d', Z(E))$. The construction packs the two-generator central detectors and the top symmetric cocycle into exactly that many center coordinates. Its proof is in Appendix G. Optimizing the center is different from retaining the full kernel of a universal central cover.

5.1 Prescribed graph quotients

Let $G = (V, \mathcal{E})$ be a finite simple graph with $|V| \geq 2$ and no isolated vertices. Define Q_G on generators x_i, $i \in V$, and basis c_e, u_e, v_e, w_e for every edge $e = \{i, j\}$, $i < j$, by

$$[x_i, x_j] = c_e, \quad [c_e, x_i] = u_e, \quad [c_e, x_j] = v_e, \qquad [u_e, x_j] = [v_e, x_i] = w_e,$$

with all other basis brackets zero. It is the ordinary metabelian three-Engel quotient of $H_{|V|}$ obtained by killing nonedge commutators and all components supported on three or more vertices. It has dimension $|V| + 4|\mathcal{E}|$, class four, and center spanned by the w_e.

Theorem 5.2 (The least center for a graph quotient)**.** *Put*

$$p(G) = \max_{\varnothing \neq W \subseteq V} \left\lceil \frac{|\mathcal{E}(G[W])|}{|W|} \right\rceil .$$

Among all finite-dimensional ordinary Lie realizations $E/Z(E) \simeq Q_G$, the minimum center dimension is $p(G)$. An attaining ordinary three-Engel algebra has exact rank $|V|$, class five, dimension $|V| + 4|\mathcal{E}| + p(G)$, and $Z(E) = E''$.

Jacobi restricts each edge-center detector to its two endpoints, so every induced subgraph gives a lower bound. The Hall–Hakimi endpoint assignment and the pair cocycles attain all these bounds simultaneously; Appendix H gives the proof. For instance $K_4 \sqcup K_2 \sqcup K_2$ has minimum two, whereas the single global center-to-Hom dimension estimate gives only one.

6 Further questions

The stable starting ranks now satisfy $a_s = \Theta(s)$. Determine their optimal asymptotic constants and exact jumps. The lower current envelope has dyadic transitions; its internal optimality does not establish the least stable rank for the whole ordinary class. Improving the jet bound requires control of the additional mixed commutator terms that appear below the $s - 1$ jet cutoff.

At the boundary rank $d = s + 2^{\lfloor \log_2(s-1) \rfloor} - 1$, determine the kernel of the symmetric-cube quotient of Theorem 4.6. The rank-eight fifth-depth layer is known completely; a quotient at the other ranks need not exhaust their free critical component. A uniform account of the preceding rank's cubic Lie structure is also open.

The optimal unrestricted growth of $e(d)$ remains a separate problem: the depth of the second derived ideal is not fixed there. For prescribed central realizations, further natural quotients may have detector constraints beyond the endpoint conditions of Q_G. Their minimum whole-center problem requires both local restrictions and a simultaneous cocycle construction.

A Proofs for the top cover and class boundary

A.1 The ordinary metabelian ceiling

We now work with ordinary Lie algebras over k and ordinary k-Lie generators. The algebra $H_d = U \oplus W_d$ is defined by (4), using exactly the module and triangle relations of Section 2.

Lemma A.1 (The classical metabelian ceiling)**.** *Every ordinary metabelian 3-Engel k-Lie algebra generated by at most $d \geq 2$ elements is a quotient of H_d. Its class is at most $\max(4, d + 1)$, and H_d attains this bound at exact ordinary generator rank d.*

Proof. Let A be metabelian, put $W = A'$, and choose a linear section of A/W. On W, write $P_u = \mathrm{ad}_u|_W$ and let $\beta(u, v)$ be the bracket of the chosen lifts. The P_u commute. The 3-Engel identity gives $P_u^3 = 0$ and $P_u^2 \beta(u, v) = 0$. The partial coefficients of $(P_u + \lambda P_v)^3 = 0$ give $P_u^2 P_v = P_u P_v^2 = 0$. The coefficient of λ in

$$P_{u+\lambda v}^2 \beta(u + \lambda v, w) = 0$$

then gives $P_u^2\beta(v,w)=0$. These coefficient separations are valid over k: the polynomials have degree at most three, strictly less than $|k|=4$. Since

$$W=\sum_v P_vW+\operatorname{span}_k\{\beta(v,w):v,w\in A/W\},$$

we obtain $P_u^2W=0$ for every u. Conversely this square-zero action makes every third adjoint zero on A.

For any chosen d generators, the actions therefore factor through $R_d=k[t_i]/(t_i^2)$. Jacobi gives precisely the triangle relations (1), so the universal commutator-module presentation supplies a surjection $H_d\to A$. For H_d,

$$\gamma_2H_d=W_d,\qquad \gamma_jH_d=\mathfrak{m}^{j-2}W_d\quad(j\geq 2).$$

Proposition 2.1 puts the highest coefficient degree at two for $d=2,3$, and at $d-1$ for $d\geq 4$. It follows that $\operatorname{cl}H_d=\max(4,d+1)$. Its abelianization is U, so its ordinary generator rank is exactly d. This is the direct class consequence of the classical commutator module. □

A.2 The complete top central cover

The exterior-coinvariant description of free centre-by-metabelian central kernels is classical; see [1, (2.1) and Theorem 7.2]. Here we first impose the square-zero action relations of W_d and then compute the full top-weight CE quotient.

Central lifts, class, and ordinary generators. For an ordinary CE cocycle $\alpha:H\wedge H\to Z$, the central extension has bracket

$$[(u,a),(v,b)]_\alpha=([u,v],\alpha(u,v)).$$

If H is 3-Engel, its extension is 3-Engel exactly when

$$\alpha([y,x,x],x)=0\qquad(x,y\in H),\tag{9}$$

because its third iterated bracket is $([y,x,x,x],\alpha([y,x,x],x))$. If H has class $c\geq 2$, projection of each lower-central term gives

$$\gamma_{c+1}(E_\alpha)=\operatorname{span}_k\alpha(\gamma_cH,H)\subseteq Z,\qquad\gamma_{c+2}(E_\alpha)=0.\tag{10}$$

For $h\in\gamma_cH\subseteq Z(H)$, the CE identity implies $\alpha(h,H')=0$; coboundaries also vanish on these top pairings. Thus a nonzero top pairing raises the class exactly once.

Let S be generated by arbitrary lifts of ordinary generators of H. Then $E_\alpha=S+Z$, and centrality gives $\gamma_jE_\alpha=\gamma_jS$ for $j\geq 2$. If $\dim Z=1$ and the top pairing is nonzero, then $Z=\gamma_{c+1}E_\alpha\subseteq S$, so $E_\alpha=S$. Its ordinary generator rank therefore equals that of H.

Proof of Theorem 3.1. Write $W=W_d$, $R=R_d$, over k. The grading used here is Lie weight: $\deg x_i=1$, $\deg v_{ij}=2$, and $\deg t_i=1$.

The top commutator space. Write $t_S=\prod_{i\in S}t_i$. The top graded part is W_{d+1}, where the subscript denotes the total Lie weight. It has a basis

$$T_i=t_{[d]\setminus\{j\}}v_{ij}\qquad(j\neq i),\tag{11}$$

and this expression does not depend on the choice of j. Indeed, the triangle relation on i,j,k, multiplied by $t_{[d]\setminus\{j,k\}}$, proves independence of j and k. If $s\notin\{i,j\}$, then

$$t_{[d]\setminus\{s\}}v_{ij}=0.\tag{12}$$

To see this choose $k \notin \{i, j, s\}$, which is possible for $d \geq 4$, and multiply the triangle relation on i, j, k by $t_{[d]\setminus\{k,s\}}$. Its first two terms have a squared variable and vanish; the third is the displayed expression. These reductions show that the T_i span the top part.

Let $\theta : R \to k$ extract the coefficient of $t_1 \cdots t_d$. The map on top-degree presentations

$$\varphi(pv_{ij}) = \theta(pt_j)x_i + \theta(pt_i)x_j \qquad (\deg p = d - 1) \tag{13}$$

annihilates each triangle relation: each resulting term occurs twice. It takes T_i to x_i, proving independence. With $\Delta = x_1 \wedge \cdots \wedge x_d$, the canonical version is

$$W_{d+1} \simeq \det(U) \otimes U, \qquad T_i \longmapsto \Delta \otimes x_i.$$

Here naturality can be checked on generators of $\mathrm{GL}(U)$. Diagonal changes have weight $\det(B)B$ on the displayed T_i. Permutations permute their indices. For $x_1 \mapsto x_1 + x_2$, (11) with $j = 2$ gives $T_1 \mapsto T_1 + T_2$, while T_2 is fixed; for $i \notin \{1, 2\}$ the extra term vanishes by (12). Thus the determinant-twisted statement is independent of the basis.

Completeness of the CE quotient. Give a central coefficient space Z weight $d + 2$. For an alternating CE cocycle $\alpha : H_d \wedge H_d \to Z$ homogeneous of total weight $d + 2$, write

$$\eta(w, u) = \alpha(w, u), \qquad \beta(w, w') = \alpha(w, w').$$

The only possibly nonzero U–W terms are $\eta : W_{d+1} \otimes U \to Z$. The CE equations on x_i, w, w' give

$$\beta(t_i w, w') = \beta(w, t_i w'), \tag{14}$$

and those on x_i, x_j, w give

$$\beta(v_{ij}, w) = \eta(t_i w, x_j) + \eta(t_j w, x_i).$$

Thus every cocycle is uniquely determined by η, with

$$\beta(rv_{ij}, w) = \eta(t_i rw, x_j) + \eta(t_j rw, x_i), \tag{15}$$

where η is zero off W_{d+1}.

Put $B_{ij} = \eta(T_i, x_j)$. Symmetry of β forces $B_{ij} = B_{ji}$. For $j \neq l$, choose i, k so that i, j, k, l are distinct, and put $r = t_{[d]\setminus\{i,k\}}$. Equations (12)–(15) give

$$\beta(v_{ij}, rv_{kl}) = B_{lj}, \qquad \beta(rv_{kl}, v_{ij}) = B_{jl}.$$

This proves the required symmetry, with no condition on the diagonal.

Conversely, let $B : U \times U \to Z$ be any symmetric bilinear map after the displayed choice of Δ, and define η by its matrix B. Define β by (15). This descends from the free R-module to W: multiplication by each triangle relation gives six η terms, in three equal pairs, because the t_i commute. The square-zero ring relations are already respected. The expression can be written explicitly as

$$\begin{aligned}\beta(rv_{ij}, sv_{kl}) = {} & \theta(rst_it_l)B_{kj} + \theta(rst_it_k)B_{lj} \\ & + \theta(rst_jt_l)B_{ki} + \theta(rst_jt_k)B_{li}.\end{aligned} \tag{16}$$

This formula is symmetric under exchanging (ij, r) and (kl, s) because B is symmetric. It is alternating. More strongly,

$$\beta(pv, v) = 0 \qquad (p \in R,\ v \in W). \tag{17}$$

For one cyclic generator $v = rv_{ij}$, the four terms in (16) reduce to two squared-variable terms, which vanish, and $\theta(pr^2 t_i t_j)(B_{ij} + B_{ji}) = 0$. For a sum of cyclic generators the cross terms cancel in pairs, using symmetry and the balanced property (14). This proves (17), and in particular alternation.

Now set $\alpha|_{U\wedge U} = 0$, use η on $W \otimes U$, and use β on $W \wedge W$. All CE equations hold: the U, U, W equations are the definition of β; the U, W, W equations are balance; the W, W, W equations vanish because W is abelian; and the U, U, U equations vanish by the homogeneous weight. Thus every symmetric B extends, uniquely, to a homogeneous CE cocycle.

There are no coboundaries in this weight: H_d has no bracket of weight $d + 2$, so $t([a, b])$ has zero component in this total weight. Consequently the homogeneous cohomology space is naturally

$$H^2_{\mathrm{CE}}(H_d, Z)_{d+2} \simeq \mathrm{Hom}_k(\det(U) \otimes \mathrm{Sym}^2 U, Z). \tag{18}$$

Automatic three-Engel admissibility. For $u, s \in U$ and $v, w \in W$, put $x = u + v$ and $y = s + w$. Write P_u for multiplication by the corresponding linear element of R, and $c(s, u) = [s, u]$. In the metabelian base,

$$[y, x, x] = P_u c(s, u) + P_u P_s v,$$

because $P_u^2 = 0$ on W. The central three-Engel obstruction is therefore

$$\begin{aligned}\alpha([y, x, x], x) = {} & \eta(P_u c(s, u), u) + \eta(P_u P_s v, u) \\ & + \beta(P_u c(s, u), v) + \beta(P_u P_s v, v).\end{aligned}$$

The first term has total weight four and vanishes because $d + 2 \geq 6$. Balance and (15) give

$$\beta(P_u c(s, u), v) = \beta(c(s, u), P_u v) = \eta(P_s P_u v, u) + \eta(P_u^2 v, s) = \eta(P_s P_u v, u).$$

It cancels the second term. The last term is zero by (17) with $p = P_u P_s$. Hence every cocycle in (18) satisfies $\alpha([y, x, x], x) = 0$ for every x, y.

Class and exact generator rank. When $\dim Z = 1$, a nonzero symmetric form has a nonzero top pairing. Equation (10) gives class $d + 2$, and the lifted generators generate the extension by the preceding central-lift argument. The ordinary rank is exactly d. The extension's second derived ideal is contained in Z. It is nonzero by Lemma A.1, since its class exceeds the metabelian ceiling $d + 1$. Hence it equals Z. □

A.3 The rank-two endpoint and the sharp boundary

Lemma A.2 (The ordinary two-generator endpoint)**.** *Over k, the free ordinary 3-Engel Lie algebra on two generators has dimension eight and class five. In particular $e(2) = 5$.*

Proof. Let F have basis a, b, c, u, v, w, z, t and nonzero unordered brackets

$$\begin{gathered}[a, b] = c, \quad [c, a] = u, \quad [c, b] = v, \quad [u, b] = [v, a] = w, \\ [w, a] = [c, u] = z, \quad [w, b] = [c, v] = t.\end{gathered} \tag{19}$$

All omitted brackets of distinct basis vectors vanish, and the bracket is alternating. In particular z, t are central. For Jacobi on distinct basis triples, the only triples with a nonzero individual nested term are $\{a, b, c\}$, $\{a, b, u\}$, and $\{a, b, v\}$. Their two nonzero terms are respectively $w + w$, $z + z$, and $t + t$. All three sums vanish. Triples with a repeated basis vector satisfy Jacobi automatically by alternation in characteristic two. This proves that the displayed table defines a Lie algebra.

For $x = Aa + Bb + Cc + Du + Ev + Fw + Gz + Tt$ and $R_x(y) = [y, x]$,

$$\begin{aligned}
R_x(a) &= Bc + Cu + Ew + Fz, & R_x(b) &= Ac + Cv + Dw + Ft,\\
R_x(c) &= Au + Bv + Dz + Et, & R_x(u) &= Bw + Cz,\\
R_x(v) &= Aw + Ct, & R_x(w) &= Az + Bt,\\
R_x(z) &= R_x(t) = 0.
\end{aligned}$$

Consequently

$$\begin{aligned}
R_x^2(a) &= ABu + B^2v + BCw + (BD + C^2 + AE)z,\\
R_x^2(b) &= A^2u + ABv + ACw + (AE + C^2 + BD)t,\\
R_x^2(c) &= ACz + BCt,\\
R_x^2(u) &= ABz + B^2t, \qquad R_x^2(v) = A^2z + ABt,\\
R_x^2(w) &= R_x^2(z) = R_x^2(t) = 0.
\end{aligned}$$

Substitution of the first table into the second gives $R_x^3 = 0$: the noncentral terms cancel in pairs, and z, t are central. Thus F is ordinary three-Engel. Its lower central dimensions are

$$(8, 6, 5, 3, 2, 0),$$

because the successive spans are F, $\langle c, u, v, w, z, t\rangle$, $\langle u, v, w, z, t\rangle$, $\langle w, z, t\rangle$, $\langle z, t\rangle$, and zero. It has exact ordinary rank two and $F'' = \langle z, t\rangle \neq 0$.

Every pair of elements has this table. Let L be any ordinary three-Engel Lie algebra over k, and fix $a, b \in L$. Define

$$c = [a, b], \quad u = [c, a], \quad v = [c, b], \quad w = [u, b] = [v, a], \quad z = [w, a], \quad t = [w, b].$$

The equality defining w is Jacobi. Engel gives

$$[u, a] = [v, b] = 0, \qquad [z, a] = [t, b] = 0.$$

For example $[z, a] = [v, a, a, a] = 0$, and the other last equality is $[u, b, b, b] = 0$. Jacobi also gives

$$[c, u] = z, \qquad [c, v] = t.$$

Set

$$p = [z, b], \qquad q = [t, a], \qquad r = [c, w], \qquad s = [u, v].$$

Jacobi on $\{c, u, b\}$, $\{c, v, a\}$, and $\{a, b, w\}$ gives

$$p + r + s = 0, \qquad q + r + s = 0, \qquad r + q + p = 0.$$

Since the characteristic is two, these imply

$$p = q, \qquad r = 0, \qquad s = p.$$

It remains to kill p. Apply the three-Engel identity to $y = a$ and $x = b + \lambda c$:

$$\begin{aligned}
0 &= [a, b + \lambda c, b + \lambda c, b + \lambda c]\\
&= [v + \lambda w + \lambda^2 z, b + \lambda c]\\
&= \lambda^2(r + p) + \lambda^3[z, c].
\end{aligned}$$

For $\lambda \neq 0$, division by λ^2 yields $r + p + \lambda[z, c] = 0$. Two distinct nonzero values of λ exist because $|k| > 2$. Subtraction first gives $[z, c] = 0$, then $r + p = 0$. As $r = 0$, it follows that $p = q = r = s = 0$.

Thus z, t commute with a, b, and are central in the subalgebra generated by a, b: the centralizer of either element is a Lie subalgebra containing the generators. Finally

$$[u, w] = [u, [v, a]] = [[u, v], a] + [v, [u, a]] = 0,$$

and similarly $[v, w] = [v, [u, b]] = [[v, u], b] + [u, [v, b]] = 0$. All unordered brackets of a, b, c, u, v, w, z, t are now exactly those in (19). Their span is a Lie subalgebra containing a, b, and hence equals $\langle a, b\rangle_L$. The linear map from F sending its two generators to this arbitrary pair is therefore a Lie homomorphism. It is unique since a, b generate F. This proves the claimed free universal property and the exact class maximum.

□

Proof of Theorem 3.2. Let L have ordinary generator rank at most $d \geq 2$, satisfy the 3-Engel identity, and have $L'' \subseteq Z(L)$. Then $L/Z(L)$ is metabelian, 3-Engel, and generated by at most d elements. Lemma A.1 gives

$$\operatorname{cl} L \leq \operatorname{cl}(L/Z(L)) + 1 \leq \max(4, d + 1) + 1 = \max(5, d + 2).$$

If the quotient is zero, the class is at most one. Otherwise the first inequality follows by lifting its lower-central series: the next term lies in $Z(L)$, and one more bracket kills it.

For $d \geq 4$, any nonzero one-dimensional top cover from Theorem 3.1 has exact ordinary rank d, class $d+2$, and central second derived ideal. For $d = 2$, the eight-dimensional algebra in Lemma A.2 has class five and central second derived ideal. Its direct sum with a central one-dimensional abelian algebra attains class five at exact rank three. These models prove sharpness at both exact rank and rank at most d. □

B A Frobenius-jet proof of uniform stability

Work over a characteristic-two field k with $|k| > 2$. Set $K = H''$ and $M_{i+1} = [M_i, H]$, with $M_0 = K$. We prove the upper bound in Theorem 4.1 independently of the finite systems used for the exact low-depth endpoints.

B.1 A partial-Engel auxiliary class

Fix r marked anchors $x_1, \ldots, x_r$ and a further letter z. For every anchor impose the formal identity

$$E_{21}(v; x_i, u) := [v, x_i, x_i, u] + [v, x_i, u, x_i] + [v, u, x_i, x_i] = 0 \quad (u, v \in H). \tag{20}$$

Also impose $M_s(H) = 0$. Let $A_{r,s}$ be the universal Lie algebra with these relations, multigraded by the anchors and z. Write n for the *total* anchor degree and m for the z-degree. Only the fixed-anchor partial identities are imposed; the letter z has unrestricted adjoint action. These partial identities are extracted from the ordinary three-Engel identity over the stated fields [7, 8].

Put $D = \operatorname{ad} z$. On $P = H'/K$ the actions of all generators commute. Modulo K, the first two terms of (20) agree, so

$$[v, u, x_i, x_i] = 0 \pmod{K}.$$

As the $[v, u]$ span H', the anchor action on P has square zero. Consequently P is a module over the genuine commutative ring

$$R = k[D, t_1, \ldots, t_r]/(t_1^2, \ldots, t_r^2), \qquad t_i = \operatorname{ad} x_i \text{ on } P. \tag{21}$$

It is generated over R by the classes of $[x_i, x_j]$ and $[x_i, z]$. Their anchor degrees are two and one. Thus

$$(A_{r,s})_{n,m} \subseteq K \quad \text{whenever } n > r + 2. \tag{22}$$

Only this elementary metabelian module is used. We do not equip K or a skew quotient of $P \otimes_R P$ with an unproved R-action.

B.2 The high-jet ideal and its square

Assume $s \geq 2$ and set $h = s-1$. Let C be the ideal generated by $D^h x_1, \ldots, D^h x_r$. This cutoff need not be a power of two. The classical Jacobi filtration estimate is

$$[M_i(H), \gamma_j(H)] \subseteq M_{i+j}(H). \tag{23}$$

It follows by induction on j from Jacobi and the fact that the M_i are ideals. In particular $\gamma_s(H)$ centralizes K.

Lemma B.1 (High jets centralize the second derived)**.** *One has*

$$C \subseteq H' \cap C_H(K), \qquad [C, C, C] = 0.$$

Every jet $D^j x_i$ with $j \geq h$ lies in C.

Proof. Each generator $D^h x_i$ belongs to $\gamma_{h+1}(H) = \gamma_s(H)$, so it centralizes K by (23). The centralizer of an ideal is an ideal, so all of C centralizes K. Since $h \geq 1$, its generators lie in H', hence $[C, C] \subseteq K$ and $[C, C, C] = 0$. Further D-actions stay in the ideal C, proving the assertion about all higher jets. □

Lemma B.2 (The square has bounded anchor degree)**.** *In the multigraded universal auxiliary algebra,*

$$[C, C]_{n,m} = 0 \qquad (n > r + 2), \tag{24}$$

at every z-degree m.

Proof. Let $V = C/(C \cap K)$, regarded as an actual R-submodule of P. By Lemma B.1, it is generated over R by the classes $v_i = D^h x_i$, each of anchor degree one. Because C centralizes K, the bracket descends to a bilinear pairing

$$\beta : V \times V \longrightarrow [C, C].$$

It is equivariant for generator actions with the *derivation* rule; balance over R is neither asserted nor needed.

For $c, d \in C$ and $x = x_i$, the metabelian calculation gives $[c, x, x], [d, x, x] \in K$. The square of $\operatorname{ad} x$ is a derivation, so $[c, d, x, x] = 0$ since C centralizes K. The following identity holds in the free Lie algebra in characteristic two:

$$[[c, x], [d, x]] = E_{21}(c; x, d) + [c, d, x, x]. \tag{25}$$

It follows that

$$\beta(vt_i, wt_i) = 0 \qquad (v, w \in V). \tag{26}$$

This is a simultaneous square relation in the two legs, rather than an unjustified scalar action on their bracket.

Expand both legs in the spanning set $D^a t_A v_i$, where $A \subseteq [r]$. A repeated anchor coefficient within one leg is zero by $t_i^2 = 0$. A coefficient repeated between the two legs gives zero by (26); commutativity of the actions on V allows the common coefficient to be placed last in each leg. Therefore the coefficient anchor sets in a surviving pair are disjoint. Their combined size is at most r, and the two v_i contribute only two further anchor occurrences. This proves (24). □

B.3 A uniform jet cone

Theorem B.3 (Fixed-anchor Frobenius-jet bound)**.** *Let $s \geq 2$ and let $p \geq s$ be a power of two. Then*

$$(A_{r,s})_{n,m} = 0 \quad \text{if } n > r+2 \text{ and } m > n(s-2)+p. \tag{27}$$

This holds in every anchor multidegree of total degree n.

Proof. Use classical Lazard elimination [9, Section 2]: the ideal generated by the anchors is free on the jets

$$x_i^{(j)} = D^j x_i, \qquad i \in [r], \quad j \geq 0.$$

For completeness, take the free Lie algebra on these jet symbols, let a derivation shift j to $j+1$, and form its semidirect product with kz. It has the universal property of the free Lie algebra on $z, x_1, \ldots, x_r$, which proves the required decomposition without any characteristic-zero assumption. A word of anchor degree n is consequently a sum of bracketed words in n jets, with the sum of their indices equal to m.

A jet with index at least $h = s-1$ lies in C. Every word with two such jets belongs to $[C, C]$: collect the ideal entries using Jacobi, or induct on the bracket tree. In anchor degree $n > r+2$ it is zero by Lemma B.2.

Next suppose a word has a jet of index $j \geq h+p$. Lower this one index by p, obtaining a word w of the same anchor degree. By (22), $w \in K$, so $D^p w = 0$ since $p \geq s$. As p is a power of two, D^p is a derivation in characteristic two. Its expansion has one term equal to the original word. In each other term the lowered jet still has index at least h, while a different jet has acquired $p \geq s > h$. Hence every other term has two entries in C and is zero. The original word is zero as well.

Thus any surviving word has at most one jet with index in $[h, h+p-1]$, and all its other indices are at most $h-1$. Its total z-degree is at most

$$(h+p-1) + (n-1)(h-1) = n(s-2)+p,$$

proving the theorem. All steps are uniform in p, r, n, s; no finite matrix rank is an input. □

B.4 Compression of a full-support critical component

The passage back to ordinary Lie algebras must retain a distinguished singleton. Identifying *all* singletons inside a free Lie algebra would incorrectly turn a bracket of two distinct singletons into $[z, z] = 0$. We instead use the component linear in a fixed singleton y, using its free cyclic-module presentation. The following surjection uses only the fixed-anchor and depth relations.

Lemma B.4 (Anchored compression of the mixed critical profiles)**.** *Suppose $d \geq 4$ and $T = (D_{d,s})_{d+3}$ has full support. Fix a weight of T, retain every repeated coordinate as an anchor, and retain one singleton y as an additional anchor. Let r be this total number of anchors, α their multidegree, and $m = d-r$ the number of remaining singletons. Then there is a vector-space epimorphism*

$$(A_{r,s})_{\alpha,m} \twoheadrightarrow T_{\alpha,1^m}. \tag{28}$$

In all three degree-$d+3$ profiles, $|\alpha| = r+3$ and $r \leq 4$. The distinguished anchor y has degree one throughout.

Proof. First, permutations of the m remaining singleton labels act trivially on the fixed weight. A shear of one singleton into another adds a term with a missing coordinate, which is zero by full support. Both opposite shears fix the weight, and their threefold product interchanges those two labels in characteristic two. Transpositions generate all singleton permutations.

The free Lie component linear in y has basis

$$[y, a_1, \dots, a_l]$$

for associative words $a_1 \cdots a_l$ in the other generators. Jacobi proves spanning. Independence can also be seen in the free associative expansion: its unique term beginning with y is $ya_1 \cdots a_l$. Replace the m occurrences of the anonymous letter z in such a word by the m distinct singleton labels in a fixed order. The resulting element of T is independent of that order by the preceding permutation argument. This defines a surjection from the free one-y component. Equivalently, this free anonymous component is the coinvariant space of the labelled associative-word permutation basis. This statement identifies basis orbits and does not average over the singleton group or divide by its order.

It remains to check the auxiliary defining relations, rather than assume a Lie identification after merging labels. Relation (20) has its repeated value equal to a fixed anchor; its other two input values each occur once. In every one-y context, allocate disjoint sets of singleton labels to those inputs and to the context. The labelled partial Engel identity is zero in the ordinary algebra. Any different order of the allocated labels has the same image in T, so the anonymous relation maps to zero. The identities with the distinguished anchor y repeated twice have y-degree at least two and cannot affect this component.

Similarly, the depth relation

$$[[u_1, u_2], [u_3, u_4], u_5, \dots, u_{s+4}] = 0$$

is multilinear in its displayed inputs. Label their singleton occurrences on disjoint sets; it is an actual M_s relation. The same argument works in every Lie-ideal context and hence kills the ideal generated by all auxiliary depth relations. Free Lie relations are already handled by the one-y associative-word presentation. In particular, when an anonymous old Lie alternation becomes a sum of two operator words, its two labelled versions are identified by a singleton permutation and cancel; no alternating power is identified with a symmetric power.

Thus the map factors through (28). The full-support weights of degree $d+3$ are $(4, 1, \dots)$, $(3, 2, 1, \dots)$, and $(2, 2, 2, 1, \dots)$. After retaining y their anchor degrees are $(4, 1)$, $(3, 2, 1)$, and $(2, 2, 2, 1)$, with $(r, |\alpha|) = (2, 5), (3, 6), (4, 7)$. □

The construction of (28) lifts every defining relation on disjoint singleton labels. It therefore applies to the entire critical weight, including all mixed profiles.

B.5 The all-depth class bound

Proof of Theorem 4.1. Theorem 3.2 and its attaining covers give $\operatorname{cl} D_{d,1} = d + 2$ for $d \geq 4$. Assume the $(s-1)$-depth law holds for every rank at least a. The map $D_{d,s} \to D_{d,s-1}$ is central, so for $d \geq a$ the only possible extra degree is $d + 3$. For $d \geq a + 1$, that critical component T has full support: any missing coordinate puts it in a $(d-1)$-generated depth-s algebra of class at most $(d-1) + 3 = d + 2$.

Choose $p = Q(s)$. Apply Lemma B.4 to each weight. In its notation $n = r + 3$ and $m = d - r$. The jet cone kills the weight as soon as

$$d - r > (r + 3)(s - 2) + p, \quad \text{that is,} \quad d \geq (r + 3)(s - 1) + p - 2.$$

Since $r \leq 4$, every mixed weight is zero for $d \geq 7s + p - 9$. Thus $T = 0$ in (7), and the class ceiling is $d + 2$. The centre-by-metabelian covers attain it at every such exact rank. This proves the recurrence and inductively proves eventual stability at every depth, without a new finite start.

The expression $7s + Q(s) - 9$ is 7 at $s = 2$ and increases by at least seven per depth. It therefore dominates the preceding bound plus one, starting from the depth-one start at most four. This proves $a_s \le 7s + Q(s) - 9$ by induction. For $s \ge 2$, $Q(s) \le 2s - 2$, whence $a_s \le 9s - 11$. Combining this with the independently proved current-family lower bound in Corollary C.4 gives $a_s = \Theta(s)$. □

C Zassenhaus currents and depth obstructions

Let k have characteristic two and $|k| > 2$. Everything below concerns ordinary Lie generators and dimensions over k. Put $q = 2^r \ge 2$ and $Q = q - 1$. The Lie input is the classical derived Zassenhaus algebra $S_q = W(1; r+1)'$ of dimension $2q - 1$. We give the presentation and all identity calculations needed here; simplicity and algebraic closure are not used. The cyclic support condition below determines its ordinary generator rank and exact central-depth filtration.

Classical differential presentation. Let O be the divided-power algebra on $z^{[a]}$, $0 \le a < 2q$, with $z^{[a]}z^{[b]} = \binom{a+b}{a} z^{[a+b]}$ and truncation at $2q$. The special derivative satisfies $\partial z^{[a]} = z^{[a-1]}$. Write $e_i = z^{[i+1]}\partial$ for $-1 \le i \le 2q - 3 = 2Q - 1$. These span S_q and

$$[e_i, e_j] = c(i,j)e_{i+j}, \qquad c(i,j) = \binom{i+j+1}{i+1} + \binom{i+j+1}{j+1} \pmod 2, \tag{29}$$

where an index outside $[-1, 2Q-1]$ gives zero. This is the standard divided-power presentation [10, Section 3, (3.2)–(3.6)]. It is a Lie algebra as a subalgebra of derivations of O. The index grading gives a cyclic grading by $\mathbb{Z}/Q\mathbb{Z}$. Set $x = e_Q$ and $z_0 = e_{-1}$ when acting on S_q. Lucas's identity gives

$$[e_i, z_0] = e_{i-1} \ (i \ge 0), \qquad [e_i, x] = \begin{cases} e_{i+Q}, & -1 \le i \le Q-1, \\ 0, & Q \le i \le 2Q-1. \end{cases} \tag{30}$$

Lemma C.1 (Differential identities behind admissibility)**.** *The adjoint matrices of S_q satisfy every six-term symmetric cubic coefficient. With $A = \operatorname{ad} e_Q$ they also satisfy*

$$A^3 = 0, \qquad A^2B + ABA + BA^2 = 0 \quad (B \in \operatorname{ad} S_q).$$

These are formal identities over the coefficient field.

Proof. For special derivations, write $\{f, g\} = fg' + f'g$. In characteristic two its derivative is $fg'' + f''g$. Therefore

$$\{\{f,g\},\{h,l\}\} + \{\{f,h\},\{l,g\}\} + \{\{f,l\},\{g,h\}\} = 0. \tag{31}$$

Indeed the expansion has derivative pattern $(0, 0, 1, 2)$ on four variables; each term is paired with the term interchanging the two undifferentiated positions. Multiplication is commutative, so they cancel. Jacobi identifies (31) with the six-term adjoint coefficient on f, g, h, applied to l.

For the repeated coefficient use $f = z^{[q]}$ and the operator $A_f(v) = fv' + f'v$ on coefficient functions. Both f^2 and $(f')^2$ are zero, since every positive divided-power element has square zero. Put $h = ff'$. A direct Leibniz expansion gives

$$A_f^2 = A_h, \qquad A_f^3 = 0, \qquad A_f A_g A_f = A_{ff'g' + ff''g} = [A_h, A_g].$$

For transparency, the coefficient of v' in $A_f A_g A_f(v)$ is $ff'g' + ff''g$; its coefficient of v is the derivative of that expression, and its higher derivative coefficients vanish. Since $[A_h, A_g] = A_f^2 A_g + A_g A_f^2$, this proves the partial coefficient. Restriction to S_q gives the assertion. □

The current family. For $m \geq 1$ put $R_m = k[t_1, \dots, t_m]/(t_i^2)$ and use t_I for square-free monomials. Define

$$J_{q,m} = kx \;\oplus \bigoplus_{\varnothing \neq I \subseteq [m]} t_I \otimes (S_q)_{-|I| \bmod Q}. \tag{32}$$

Brackets use disjoint coefficient supports, and x acts by $\operatorname{ad} e_Q$. Equivalently this is a subalgebra of $S_q \otimes R_m$ with $x = e_Q \otimes 1$. The marked generators are $x, z_i = t_i e_{-1}$, $1 \leq i \leq m$. The cyclic grading, rather than a full tensor product with every copy of S_q, is essential for the generator rank.

Theorem C.2 (Exact two-parameter class law)**.** *The algebra $J_{q,m}$ is ordinary three-Engel, has exact rank $d = m + 1$, and has a one-dimensional center. Put*

$$c(q, m) = m + \left\lfloor \frac{m-1}{q-1} \right\rfloor + 2.$$

Then

$$\operatorname{cl} J_{q,m} = c(q, m), \tag{33}$$

$$M_u(J_{q,m}) = \gamma_{q+u+3}(J_{q,m}) \quad (u \geq 0), \tag{34}$$

$$\operatorname{cl}(J_{q,m}/M_s) = \min\bigl(c(q, m), q + s + 2\bigr) \quad (s \geq 0). \tag{35}$$

Here $M_0 = J''_{q,m}$, $M_{u+1} = [M_u, J_{q,m}]$. All quotients retain ordinary rank $m+1$. The dimension is

$$\dim J_{q,m} = 2^{m+1} - 1 + \sum_{\substack{1 \leq b \leq m \\ b \equiv 1 \pmod{q-1}}} \binom{m}{b}.$$

In particular its exact second-derived central depth is $\max(0, c(q, m) - q - 2)$.

Proof. Engel and generation. A positive support coefficient squares to zero. Terms repeating the same such coefficient vanish, and disjoint coefficients carrying the same anonymous Lie value cancel in pairs in a six-term coefficient. The remaining coefficients are Lemma C.1; the only zero-support basis value is x. Thus the complete ordinary Engel identity holds, including all formal coefficients.

For a singleton support the successive x actions on z_i give $e_{-1}, e_{Q-1}, e_{2Q-1}$, the complete cyclic component of residue -1. Induct on support size. Bracketing a full preceding cyclic component with a new z_i differentiates its indices. Its image contains every target index except possibly $2Q - 1$. That exceptional index occurs only in residue -1 and is obtained by applying x to e_{Q-1}. Every required support set can be built by partitioning off one label. This proves generation without factorials or averaging.

For a current basis element $t_I e_j$, $b = |I|$, assign the integer degree

$$\deg(t_I e_j) = b + \frac{b + j}{Q}, \qquad \deg x = 1. \tag{36}$$

The generators have degree one and brackets add degrees. The degree-one space is exactly $kx \oplus \operatorname{span}\{z_i\}$, so the ordinary rank is $m + 1$. Generation in degree one also gives $\gamma_n = \bigoplus_{g \geq n} (J_{q,m})_g$. At support b, the largest possible $(b+j)/Q$ is $\lfloor (b-1)/Q \rfloor + 2$, and the resulting maximum increases with b. This proves (33).

The complete second-derived boundary. We claim that $J''_{q,m}$ consists precisely of the top index e_{2Q-1} on supports of size q, together with every cyclic component on supports of size at least $q + 1$. Empty ranges mean zero.

Since the algebra is generated in degree one, its derived ideal is all degrees at least two. For supports of sizes $b_1, b_2 > 0$ with $b = b_1 + b_2 \leq q$, derived inputs have indices $Q - b_i$ or

$2Q - b_i$; the index -1 on a singleton was excluded by the derived condition. If both are lower choices and $b \le Q$, the bracket coefficient is

$$\binom{2q-1-b}{q-b_1} + \binom{2q-1-b}{q-b_2} = 0.$$

Both lower arguments exceed the lower binary block $q-1-b$ of the upper argument, so Lucas gives zero termwise. At $b = q$ both coefficients are one and cancel. If one index is an upper choice, the output index is at least $3Q - b$; it is outside S_q for $b \le Q$, and at $b = q$ its only possible value is $2Q - 1$. Two upper choices are outside. Thus no second-derived value occurs before support q, and that support has only the specified top line.

This top line is attained: bracket e_{2Q-1} on one singleton with e_0 on a disjoint support of size Q. On support size $q+1$, the bracket of e_{-1} at size q with e_{Q-1} at size one gives e_{Q-2}; differentiating the preceding top line gives e_{2Q-2}. These give the complete cyclic component. For $Q = 1$, its remaining top index follows by an x action. The same generation argument by differentiation and x propagates full components to all larger supports, since the second derived is an ideal. All operations use fixed disjoint label sets.

By (36) the specified subspace is exactly the sum of all degrees at least $q+3$: at support q only the top line has that degree, and at larger supports every degree is at least $q+3$. Consequently $M_0 = \gamma_{q+3}$ and successive brackets give (34). Equation (35) follows, including its nonzero top degree, from degree-one generation. The truncation ideal has degree at least $q+3$, so the degree-one space is preserved.

Center and dimension. A central element has no x coefficient, by looking at its bracket with z_i in singleton support. If a current coefficient has a proper support I, choose a missing label. Its bracket with that z_i forces its S_q value into $\ker \operatorname{ad} e_{-1} = ke_{-1}$. Its bracket with x then forces it to zero. Different supports cannot cancel in this test: for fixed missing i, $I \mapsto I \cup \{i\}$ is injective. On full support only the x bracket remains. The kernel of $\operatorname{ad} x$ is spanned by $e_Q, \dots, e_{2Q-1}$, and each cyclic component meets this interval in exactly one line. This gives the one-dimensional center, in the largest homogeneous degree. Every cyclic component has dimension two, with one additional dimension when $b \equiv 1 \pmod{Q}$. This also holds for $Q = 1$, when every component has dimension three. Counting supports gives the displayed dimension. □

Recursion in the support variables. Adding a label gives the explicit semidirect product

$$J_{q,m+1} = J_{q,m} \ltimes V_{q,m}, \qquad V_{q,m} = \bigoplus_{I \subseteq [m]} t_{m+1} t_I (S_q)_{-(|I|+1) \bmod Q}.$$

The new ideal is abelian and generated as a cyclic module by $t_{m+1}e_{-1}$. At $q = 2$ this is the classical three-dimensional disjoint-support current construction with a standard inner derivation [7, 6]. For every q, the preceding calculation gives ordinary generation and exact depth directly.

Lemma C.3 (Affine changes of the outside generator)**.** *For every linear coefficient $\ell = \sum a_i t_i$, there is a graded automorphism of $J_{q,m}$ sending x to $x + \sum a_i z_i$ and fixing all z_i. All degree-one hyperplanes except* $\operatorname{span}(z_1, \dots, z_m)$ *therefore generate subalgebras conjugate to $J_{q,m-1}$, for $m \ge 2$.*

Proof. In $O \otimes R_m$, use binary divided-power generators $u_j = z^{[2^j]}$, $0 \le j \le r$, where $u_r = z^{[q]}$. Each has square zero, and $\partial u_j = \prod_{i<j} u_i$. The substitution $u_r \mapsto u_r + \ell$, fixing all lower u_i, is an involutive algebra automorphism commuting with ∂. On special derivations it sends

$$e_i \longmapsto \begin{cases} e_i, & i < q-1, \\ e_i + \ell e_{i-q}, & i \ge q-1. \end{cases}$$

It preserves the cyclic current condition: adding a coefficient label changes b by one and changes the index by $-q \equiv -1 \pmod{Q}$. It preserves total degree in (36), sends x to $x+\ell e_{-1}$, and fixes z_i.

For a hyperplane W not equal to the commuting z-space, choose $x+v \in W$. The inverse shear sends it to a hyperplane containing x. Its intersection with the z-space has codimension one; a coefficient GL_m change sends that intersection to the first $m-1$ singleton directions. The generated subalgebra is $J_{q,m-1}$ by the support presentation and generation already proved. □

Corollary C.4 (A necessary bound at every depth)**.** *For $s \geq 3$ put $q_s = 2^{\lfloor \log_2(s-1) \rfloor}$. Then the stable starting rank of Theorem 4.1 satisfies*

$$a_s \geq s + q_s \geq \left\lceil \frac{3s}{2} \right\rceil .$$

The witnessing ordinary three-Engel algebra can always be chosen with one-dimensional center, exact rank $s+q_s-1$, class $s+q_s+2$, and exact second-derived central depth s.

Proof. More generally let q be any power of two with $q \leq s-1$, and set $m = q+s-2$. Since $m \geq 2q-1$, one has $c(q,m) \geq m+4$. Equation (35) gives the quotient by M_s exact class $q+s+2 = m+4 = d+3$, proving $a_s \geq s+q$. Choose the largest available power q_s; then $q_s+1 \leq s \leq 2q_s$.

If $s < 2q_s$, the full current already has class $m+4$ and exact depth s, so its center is one-dimensional by the theorem. If $s = 2q_s$, then $m = 3q_s - 2 = 3Q+1$, the full current has class $m+5$, and M_s is its central top line $t_{[m]}e_{2Q-1}$. In the quotient the full-support line $t_{[m]}e_{Q-1}$ becomes central. No proper-support line becomes central: differentiating a proper-support value could land in the killed line only from index $2Q$, which does not exist. The x test then excludes all proper supports as before. The quotient center is therefore exactly this one line. The top surviving degree is $q+s+2$, so the exact depth is s in both cases. □

Corollary C.5 (The witnesses are critical for their class)**.** *Every proper subalgebra and every proper quotient of each witness in Corollary C.4 has strictly smaller nilpotency class.*

Proof. Let its class be $N = m+4$. Lemma C.3 and the class formula show that every proper degree-one subspace generates class below N; indeed $c(q, m-1) < N$ throughout $q+1 \leq s \leq 2q$, including the one-line quotient at the last endpoint. A proper subalgebra has proper image in the degree-one abelianization: otherwise the standard nilpotent generation argument gives the whole algebra. A word of length N in it depends only on these degree-one images, since any higher-degree input exceeds the maximal degree. Thus the word vanishes. Every nonzero ideal of a finite nilpotent Lie algebra meets its center. The center here is the one-dimensional top line, so every proper quotient kills that line and lowers class. □

Corollary C.6 (Surviving outer commutator errors)**.** *Let $q = q_s$, $k = s-q-1$, and take the witness of the preceding corollary. Choose disjoint labels $S \cup \{j\}$ of size $2q-1$, with $|S| = 2q-2$. Put $v = t_S e_Q$, $a = x$, and $b = t_j e_{-1}$. Then $v \in M_{q-2}$, $[v,a,b] = 0$, and*

$$[v,b,a] = [v,[a,b]] = t_{S\cup\{j\}} e_{2q-3} \neq 0.$$

Appending the k remaining singleton actions gives a nonzero central value in M_{s-1}/M_s, of index $2q-3-k$.

Proof. The index actions in (30) give the displayed values. The degree of v is $2q+1$, which is the first degree of M_{q-2}. Each new singleton differentiates the index once. Since $0 \leq k \leq q-1$, the resulting index remains nonnegative; its degree is $q+s+2$, the final class. The depth count is $(q-2)+2+k = s-1$. Thus the outer exchange errors are retained by the construction, including at arbitrarily large depths. □

Corollary C.7 (An all-rank lifting bound)**.** *For $d \geq 4$ let q_d be the largest power of two at most $d/2$. For ordinary central lifts over the complete bottom $E_d = D_{d,1}$,*

$$R(d) \leq d - q_d < \frac{3d}{4}.$$

In particular $R(10) = 6$.

Proof. Take $m = d - 1$, $q = q_d$, and $s = d - q + 1$. Then $m \geq 2q - 1$ and $J_{q,m}/M_s$ has exact rank d, depth s, and class $d + 3$. Write this quotient as B. Let A be the subalgebra of $B \times E_d$ generated by the diagonal marked generators. Projection onto B is surjective and $M_1(E_d) = 0$, so for every $j \geq 1$,

$$M_j(A) = M_j(B) \times 0.$$

The kernel of the projection onto E_d is exactly $M_1(A)$: the defining kernel of the free map onto E_d is $M_1(\mathcal{F}_d)$, whose image in B is $M_1(B)$. Hence $A/M_1(A) = E_d$. The quotients $A/M_j(A)$ give $s - 1 = d - q$ central lifts from this complete bottom to an algebra of class $d + 3$. Since $q_d > d/4$, the strict bound follows. For $d = 10$, $q = 4, m = 9, s = 7$ gives the attaining six lifts, whereas the sixth-depth stability in Theorem 4.4 excludes five or fewer. □

The exact obstruction envelope of this family. For fixed s, a truncated current of rank $d = m + 1$ has class greater than $d + 2$ exactly when there is a power of two q satisfying

$$2q \leq d \leq s + q - 1.$$

Consequently the largest such rank within the entire two-parameter family is $s + q_s - 1$. This is optimality within this explicit family, not a global upper bound or a proof that $a_s = s + q_s$. Together with Theorem 4.1, these bounds give $\liminf a_s/s \geq 3/2$ and $\limsup a_s/s \geq 2$.

For the full ordinary variety, if $f_s(d) = \operatorname{cl} D_{d,s}$ and $d \geq 4$, the constructions give the explicit lower envelope

$$f_s(d) \geq \max\left\{ d + 2,\ \sup_{q=2^r \geq 2} \min\left(d + \left\lfloor \frac{d-2}{q-1} \right\rfloor + 1,\ q + s + 2 \right) \right\}.$$

The $q = 2$ part recovers the classical current truncation

$$\operatorname{cl}(J_{2,m}/M_s) = \min(2m + 1, s + 4). \tag{37}$$

These controlled-depth comparisons do not establish optimal unrestricted growth of $e(d)$.

D Uniform symmetric-cube quotients

We prove Theorem 4.6 using the actual current realizations and the usual polynomial cubic representations [4, Section 12]. Let $s \geq 3$, $q = 2^{\lfloor \log_2(s-1) \rfloor}$, $d = s + q - 1$, $m = d - 1$, $k_0 = s - q - 1$, and $N = d + 3$. Let $U = k^d$, where $\operatorname{char} k = 2$ and $|k| > 2$. Use the ordinary free controlled-depth algebra $D_{d,s}$ of Section 4. All group actions are algebraic, defined by the common formal Engel coefficient presentation, and commute with field extension.

The central functional. Use the one-dimensional-center witness $E_{q,s} = J_{q,m}/M_s$ of Corollary C.4, on marked generators $X, z_1, \ldots, z_m$. Its top homogeneous degree N is the line $t_{[m]} e_{2q-3-k_0}$, of marked weight $(4, 1, \ldots, 1)$. Let λ extract that coefficient after evaluation in $E_{q,s}$.

Lemma D.1 (Rank-essential top functional)**.** *The functional λ vanishes on all degree-N Lie words in any proper linear subspace of the degree-one generator space.*

Proof. It suffices to work over an algebraic closure and test hyperplanes. The affine automorphisms of Lemma C.3 send X to $X + \sum a_i z_i$ and fix all z_i; coefficient GL_m changes are also automorphisms. They preserve M_s since it is the specified lower central term. Every hyperplane except the commuting z-space therefore generates an image of $J_{q,m-1}$. Its class is $c(q, m-1) < m + 4 = N$ throughout $q + 1 \le s \le 2q$, including the last one-line central quotient. The commuting hyperplane gives zero as well. Thus λ vanishes on every proper degree-one subspace.

□

The algebraic orbit quotient. Let $T = (D_{d,s})_N$. Let $M \subseteq T^*$ be the algebraic $\mathrm{GL}(U)$-span of λ and put $\mathcal{T} = M^*$, a quotient of T. Equivalently, quotient T by the common kernel of all formally transformed top functionals. Lemma D.1 makes every weight of $\mathcal{T}$ have full support. Therefore

$$P = \det(U)^{-1} \otimes \mathcal{T}$$

is polynomial homogeneous of degree three: its integral weights are nonnegative and sum to three. To recall the elementary polynomiality criterion, a rational matrix coefficient lies in $k[g_{ij}, \det(g)^{-1}]$. Test a generic rank-one degeneration $g\,\mathrm{diag}(t, 1, \dots, 1)h$. Nonnegative diagonal weights exclude a pole at the generic point of the determinant divisor; unique factorization removes every determinant denominator. This uses integral weights over the algebraic closure, not finite-field diagonal characters.

Lemma D.2 (Cubic orbit bound)**.** *The polynomial representation P embeds in* $\mathrm{Sym}^3 U$.

Proof. View $P^* = \det(U) \otimes M$ as a polynomial degree-three representation in the dual standard space U^*. It is generated by the transformed functional of pure weight three. That vector is fixed by the stabilizer of a chosen nonzero vector in U^*: roots inside the complementary block and roots preserving its line would lead to a negative weight, and the complementary diagonal weights are zero. Scaling the chosen vector by c scales the functional by c^3.

Consequently its orbit gives a well-defined regular homogeneous map of degree three from $U^* \setminus \{0\}$ to P^*. It extends to a polynomial map on U^*: on the coordinate chart $v_i \neq 0$ choose an invertible matrix whose first column is v; the resulting coordinate functions lie in $k[v, 1/v_i]$, and their intersection over all charts is $k[v]$ since $d \ge 2$. The representing property of divided powers then gives a surjection $\Gamma^3(U^*) \twoheadrightarrow P^*$. Its image spans the orbit by construction. Duality gives the asserted injection into $\mathrm{Sym}^3 U$. This is the elementary degree-three version of the classical highest-weight orbit argument. □

Lemma D.3 (The distinct cubic weight survives)**.** *The residual weight $(1, 1, 1)$ of P is nonzero.*

Proof. The ordinary word

$$D_{q,s} = [X, z_1, X, z_2, \dots, z_q, X, z_{q+1}, \dots, z_{2q-1}, X, z_{2q}, \dots, z_m]$$

has $\lambda(D_{q,s}) = 1$ by the current bracket table and its k_0 successive differentiations. Replace its second and third occurrences of X by z_1, z_2 to form

$$W_{q,s} = [X, z_1, z_1, z_2, z_3, \dots, z_q, z_2, z_{q+1}, \dots, z_{2q-1}, X, z_{2q}, \dots, z_m].$$

Apply the two root substitutions $z_1 \mapsto z_1 + tX$ and $z_2 \mapsto z_2 + uX$. The tu coefficient of the transformed functional on $W_{q,s}$ has four terms. Replacing the first z_1 gives the initial bracket

$[X, X] = 0$, accounting for two terms. Replacing the second z_1 and the first z_2 gives the prefix $[X, z_1, X, X] = 0$ by the ordinary three-Engel identity. The remaining choice is $D_{q,s}$ and has value one. Thus the four terms are $(0, 0, 0, 1)$.

The original weight of $W_{q,s}$ is $(2, 2, 2, 1, \dots, 1)$, so after subtracting the determinant this is exactly the distinct cubic weight. The root coefficient proves the claim in the actual Lie module. □

Proof of Theorem 4.6. Inside $\operatorname{Sym}^3 U$ the repeated-monomial submodule $L = \operatorname{span}\{u_i^2 u_j\}$ is $U^{[1]} \otimes U$, and the quotient is $\Lambda^3 U$. Both are simple for the algebraic group, as can be seen directly from their one-dimensional weight spaces. In L, a shear sends any repeated weight to a pure cube, and a pure cube generates all repeated weights by formal root coefficients and permutations. In the exterior quotient permutations generate all three-subset weights.

The extension does not split when $d \geq 3$. A weight-respecting section must send $u_1 \wedge u_2 \wedge u_3$ to $u_1 u_2 u_3$, but the shear $u_3 \mapsto u_3 + u_1$ fixes the former and adds $u_1^2 u_2$ to the latter. It follows that the only submodules of $\operatorname{Sym}^3 U$ are 0, L, and the whole module: any submodule mapping nontrivially to the simple exterior quotient either contains L or splits the extension.

By Lemmas D.2 and D.3, P is a submodule with a distinct weight, so it is the whole symmetric cube. This proves the equivariant quotient and its dimension.

For an actual covariant marked algebra, start with the class-N truncation of $D_{d,s}$ and intersect the kernels of all formally changed markings into $E_{q,s}$. This is a homogeneous invariant Lie ideal. Its degree-N annihilator is exactly M, so the quotient's top layer is $\mathcal{T}$, so it has class N. The identity marking induces a surjection onto $E_{q,s}$; hence its depth is at least s, and $M_s = 0$ gives equality. The same surjection preserves the d independent degree-one generators, proving the exact generator rank. The construction is defined over the common formal coefficient field $\mathbb{F}_2$; all arguments descend to every stated k. Algebraic orbit coefficients, rather than the finite set $\mathrm{GL}_d(k)$ alone, are used in this definition when k is finite. □

E Central-depth bounds and their exact starting systems

This appendix proves Theorem 4.2 and Proposition 4.3. Work over a characteristic-two field k with $|k| > 2$ and use left-normed brackets. Set $K = H''$, $M_0 = K$ and $M_{i+1} = [M_i, H]$. Write $F_d = \mathcal{F}_d$ for the relatively free ordinary three-Engel algebra on d degree-one generators, as in Section 4. For $U = k^d$, use

$$R = \Lambda(U), \qquad P_d = \langle c_{ij} = c_{ji},\ c_{ii} = 0 \mid t_i c_{jk} + t_j c_{ik} + t_k c_{ij} = 0 \rangle_R, \qquad B_d = \operatorname{Sym}^2_R(P_d).$$

The grading is $\deg t_i = 1$, $\deg c_{ij} = 2$. Write $P = P_d$ and $B = B_d$ when the rank is fixed. The tensor product defining B_d is over R; it is distinct from a tensor product over k carrying a diagonal derivation action. The arguments use the classical commutator and Koszul constructions [2, 1, 3].

E.1 The metabelian presentation and its top

In a metabelian three-Engel algebra, the actions on the derived ideal commute. The coefficient of two copies of x and one copy of z in $[y, x + \lambda z, x + \lambda z, x + \lambda z] = 0$ gives

$$[y, z, x, x] + [y, x, z, x] + [y, x, x, z] = 0.$$

The last two terms agree modulo H'', so $t_x^2(H'/H'') = 0$. Jacobi supplies the displayed triangle relation. Conversely $U \oplus P$ with $[u, v] = c(u, v)$ and $[w, u] = t_u w$ is metabelian and

three-Engel: after the first bracket, the next two actions give $t_x^2 = 0$. Thus this is the free metabelian object in question.

For $d \geq 4$, $P_n = 0$ for $n > d+1$. The only possible coefficient in degree $d+2$ is $t_1 \cdots t_d$, and it kills every c_{ij}: multiply the triangle on i, j, k by $t_i t_j$ for a distinct k, then multiply by the remaining variables. Lower-support cases are covered by the same argument and the degree bound for the active ring. The degree-$d+1$ elements reduce to

$$T_i = t_{[d]\setminus\{j\}} c_{ij} \quad (j \neq i).$$

Triangles identify the choices of j, and

$$\iota : P \to R \otimes U, \qquad c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i$$

takes T_i to $(t_1 \cdots t_d) \otimes x_i$. Therefore $P_{d+1} \cong \det(U) \otimes U$ and the metabelian class ceiling is $d+1$ in these ranks. This argument also shows the finite presentation being used. The argument proves injectivity on the displayed graded top and does not require global injectivity.

E.2 The top symmetric square of the Koszul presentation

The following module statement has a direct, rank-uniform proof:

$$(B_d)_{d+2} \cong \det U \otimes \operatorname{Sym}^2 U \quad (d \geq 5), \qquad (B_d)_n = 0 \quad (n > d+2,\ d \geq 4). \tag{38}$$

For the second assertion, degree $d+3$ suffices since B_d is generated in degree four over R. Proceed by induction on the active support. Supports of size at most three have no degree above six: in rank three, a coefficient using all three variables kills every c_{ij} by in the presentation of P_d; ranks one and two are immediate. For active support $r \geq 4$, every exponent in a surviving symbol is at most three. In degree $r+3$ only the patterns $(3, 2, 1, \ldots, 1)$ and $(2, 2, 2, 1, \ldots, 1)$ remain.

For the first pattern, with repeated indices i, j, the symbols are

$$D = c_{ij} \odot c_{ij}\, t_{[r]\setminus\{j\}}, \qquad X_k = c_{ij} \odot c_{ik}\, t_{[r]\setminus\{k\}} \quad (k \notin \{i, j\}).$$

Triangle (i, j, k) paired with c_{ij} gives $X_k = D$. Triangle (i, j, k) paired with c_{il}, with four distinct indices, gives $X_l = 0$: its other two terms have squared coefficients. Thus all these symbols vanish.

For the second pattern, with repeated indices i, j, k, the symbols are

$$A_i = c_{ij} \odot c_{ik}\, t_{[r]\setminus\{i\}} \quad \text{and cyclic variants,}$$

$$B_{i,a} = c_{ia} \odot c_{jk}\, t_{[r]\setminus\{a\}} \quad (a \notin \{i, j, k\}),$$

with cyclic variants. Triangle (i, j, k) paired with c_{ia} gives $B_{i,a} = 0$; triangle (i, j, a) paired with c_{ik} gives $B_{j,a} = A_i$. A spare fourth index exists for $r \geq 4$, so all symbols vanish. This completes the induction and the upper-degree bound.

For degree $d+2$ and $d \geq 5$, that bound also shows that every component with fewer than d active variables is zero. The full-support patterns are precisely $(3, 1, \ldots, 1)$ and $(2, 2, 1, \ldots, 1)$.

For the first pattern fix the tripled index i. Its symbols are

$$X_{jk} = c_{ij} \odot c_{ik}\, t_{[d]\setminus\{j,k\}} \quad (j, k \neq i,\ j \neq k).$$

Triangle (i, j, k) paired with c_{ih}, for distinct $j, k, h \neq i$, gives $X_{kh} = X_{jh}$. All symbols are therefore equal.

For the second pattern fix the doubled indices i, j and put $O = [d] \setminus \{i, j\}$. The symbols are

$$\begin{aligned} D &= c_{ij} \odot c_{ij}\, t_O, \\ A_a &= c_{ij} \odot c_{ia}\, t_{[d]\setminus\{i,a\}}, & B_a &= c_{ij} \odot c_{ja}\, t_{[d]\setminus\{j,a\}}, \\ C_{ab} &= c_{ia} \odot c_{jb}\, t_{[d]\setminus\{a,b\}}, & E_{ab} &= c_{ij} \odot c_{ab}\, t_{[d]\setminus\{a,b\}}, \end{aligned}$$

where $a, b \in O$, $a \neq b$, and $E_{ab} = E_{ba}$. Triangles give

$$D = A_a + B_a, \qquad C_{ab} = A_a = B_b, \qquad E_{ab} = A_a + A_b.$$

With three distinct indices $a, b, c \in O$, triangle (i, a, b) paired with c_{jc} gives $C_{bc} = C_{ac}$, hence $A_b = A_a$. Since $|O| \geq 3$, all A_a, B_a, C_{ab} have one common value and $D = E_{ab} = 0$. Each full-support component has dimension at most one.

To prove nonvanishing and naturality, use the Koszul map

$$\iota : P_d \to R \otimes U, \qquad c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i.$$

Multiplication defines

$$\mu : \mathrm{Sym}^2_R(P_d) \to R \otimes \mathrm{Sym}^2 U.$$

In degree $d+2$ its target is $R_d \otimes \mathrm{Sym}^2 U = \det U \otimes \mathrm{Sym}^2 U$. The X_{jk} map to $(t_1 \cdots t_d) \otimes x_i^2$ and the A_a map to $(t_1 \cdots t_d) \otimes x_i x_j$. This proves (38), including the natural module identification. In characteristic two the symmetric square here is not silently identified with a space of divided powers.

E.3 Peeling an operator into the first outer input

Let $1 \leq r \leq 3$, and assume $M_{r+1} = 0$. Jacobi gives $[M_i, H'] \subseteq M_{i+2}$, so

$$\Theta_r(w, v; a_1, \ldots, a_r) = [\widetilde{w}, \widetilde{v}, a_1, \ldots, a_r]$$

is well defined on $(H'/K)^2 \times U^r$. It is balanced in the two derived legs. The partial Engel identity with inputs a, a, v, followed by $a_2, \ldots, a_r$, gives

$$\Theta_r(t_a w, v; a, a_2, \ldots, a_r) = 0.$$

Indeed its other terms belong to M_{r+1}: the term involving $[w, a, a] \in K$ first brackets with $v \in H'$, landing in M_2, and then has $r - 1$ outer actions. Polarization exchanges an operator with the first outer input.

Consequently, for $N \geq r + 5$, there is a natural epimorphism

$$B_{N-r+1} \otimes U^{\otimes(r-1)} \twoheadrightarrow (M_r)_N. \tag{39}$$

For a square-free coefficient t_S, peel one factor $a \in S$ and send a symbol to

$$t_S(c_{ij} \odot c_{kl}) \otimes a_2 \otimes \cdots \otimes a_r \longmapsto \Theta_r(t_{S\setminus\{a\}} c_{ij}, c_{kl}; a, a_2, \ldots, a_r).$$

The exchange makes the choice independent, kills squared coefficients, and respects each triangle relation by peeling from its common coefficient, of degree $N - r - 4 \geq 1$. This proves a quotient map; it does not put an unproved residual R-action on an exterior quotient.

The second-layer ceiling. In the free graded algebra with $M_2 = 0$, the quotient by M_1 is centre-by-metabelian. Its quotient by the centre is metabelian and has class at most $d+1$, hence it has class at most $d+2$. Every homogeneous part in degree $N > d+2$ is therefore in M_1. But (39) for $r = 1$ is an epimorphism $B_N \twoheadrightarrow (M_1)_N$, and (38) kills B_N. Thus the free depth-two object already has class at most $d+2$ for every $d \geq 4$. This elementary proof needs neither the full top central cohomology classification nor an external unrestricted Engel class bound.

E.4 An explicit collecting identity for the third layer

Work in characteristic two over a field with more than three elements. Brackets are left normed. Set $v = [p, a]$, $b = [p, q]$, and define

$$E_{21}(y; x, z) = [y, x, x, z] + [y, x, z, x] + [y, z, x, x],$$

$$E_{111}(y; x, z, t) = \sum_{\sigma \in S_3} [y, x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}], \qquad (x_1, x_2, x_3) = (x, z, t).$$

Both vanish in an ordinary three-Engel algebra: they are the complete cubic coefficients, whose extraction is valid because $3 < |k|$. The permutations in the second formula are indexed permutations, even when some substituted values happen to coincide.

Write

$$D(x_1, x_2, x_3, x_4; x_5, x_6, x_7) = [[[x_1, x_2], [x_3, x_4]], x_5, x_6, x_7].$$

Let $D_1, \ldots, D_{13}$ be the following displayed values or brackets of values. Each is zero if $[H'', H, H, H] = 0$.

1	$D(p, a, [p, q], q; h, r, r)$
2	$D(p, a, [p, q], q; r, r, h)$
3	$[D(p, q, p, a; r, r, q), h]$
4	$[D(p, a, h, r; p, q, q), r]$
5	$[D(p, a, h, r; q, q, p), r]$
6	$D([p, q], q, h, r; r, p, a)$
7	$D([p, q], q, h, r; r, a, p)$
8	$D(h, r, [p, a], r; p, q, q)$
9	$D(h, r, [p, a], r; q, q, p)$
10	$D(q, r, [p, q], r; p, a, h)$
11	$D(q, r, [p, q], r; a, p, h)$
12	$D(p, a, [p, q], r; q, r, h)$
13	$D(p, q, q, r; [p, a], r, h)$

The following equality holds already in the free ordinary Lie algebra in characteristic two:

$$[[[p, q], q, r], [p, a], h, r] = [E_{21}(v; r, b), q, h] + [E_{111}(r; q, v, [b, r]), h] + \sum_{i=1}^{13} D_i. \tag{40}$$

It is an explicit collecting identity, using only the Lie relations. It can be checked by expanding commutators; `verification/central_depth/verify_sparse_seed.py` does this independently, without a quotient-space or rank calculation. Under the three-Engel and depth-three hypotheses every summand on the right vanishes, proving the seed.

E.5 The third-layer source and its vanishing

In the free depth-three object put $T_d = (M_2(F_d)/M_3(F_d))_{d+3}$. The preceding module calculation and peeling construction give a natural epimorphism

$$\Psi_d : (\det U \otimes \operatorname{Sym}^2 U) \otimes U \twoheadrightarrow T_d, \qquad U = k^d, \quad d \geq 5. \tag{41}$$

Its ingredients are the metabelian presentation over $R = \Lambda(U)$, the central pairing

$$\Theta(w, v; a, b) = [\widetilde{w}, \widetilde{v}, a, b],$$

balance $\Theta(t_a w, v; h, z) = \Theta(w, t_a v; h, z)$, and the exchange

$$\Theta(t_a w, v; h, z) = \Theta(t_h w, v; a, z).$$

It identifies the degree-$d+2$ part of $\mathrm{Sym}^2_R(P_d)$ with $\det U \otimes \mathrm{Sym}^2 U$ via the Koszul multiplication

$$c_{ij} \longmapsto t_i \otimes x_j + t_j \otimes x_i.$$

Only the source in (41) is a symmetric tensor module: T_d is a quotient, whose extra relations must still be proved.

Fix distinct coordinate indices p, q, a, h, and put $S = [d] \setminus \{p, a\}$, with $h \in S \setminus \{q\}$. For any coordinate generator z, the word

$$P^{(d)}_{pq,z} = \Theta(t_{S\setminus\{h\}} c_{pq}, c_{pa}; h, z) \tag{42}$$

is the image of $(t_1 \cdots t_d) \otimes x_p x_q \otimes z$ under (41). This follows directly by multiplying the two Koszul images.

The rank-five case. Equation (40) gives the universal identity

$$[[[p, q], q, r], [p, a], h, r] = 0. \tag{43}$$

It holds in every ordinary three-Engel algebra with $M_3 = 0$. The five formal variables are p, q, a, h, r. The multidegree is $(2, 2, 1, 1, 2)$. The explicit identity above and its free-associative expansion certify all its Lie inputs.

For $d = 5$, (43) is exactly (42) with $z = r$ distinct from p, q. It kills the image of one tensor $x_p x_q \otimes x_r$ with three distinct indices. Permutation matrices kill all such coordinate tensors. A transvection $x_r \mapsto x_r + x_p$, fixing x_p, x_q, then kills $x_p x_q \otimes x_p$; similarly it kills the tensor with last input x_q. Hence every off-diagonal tensor $x_p x_q \otimes z$ is killed.

Finally, for any q, z, choose $p \notin \{q, z\}$. The transvection $x_p \mapsto x_p + x_q$ fixes z and has determinant one. Applied to the already killed tensor $x_p x_q \otimes z$, it kills $x_q^2 \otimes z$. Therefore the entire source of (41) is killed, and $T_5 = 0$.

Propagation to arbitrary rank. Assume $T_d = 0$ for some $d \geq 5$. Add a new generator y. For old p, q, z, choose a, h as in (42). Since (42) is zero in the relatively free depth-three algebra on the old generators, it remains zero after every substitution

$$x_p \longmapsto x_p + \lambda[x_p, y], \qquad \lambda \in k.$$

There are two occurrences of x_p in (42), or three if $z = x_p$. Thus this is a vector-valued polynomial in λ of degree at most three. A polynomial of degree at most three vanishing on all four elements of $\mathbb{F}_4$ has all coefficients zero. In particular its coefficient of λ vanishes. The field-size check is necessary here; an unqualified extraction from a degree-four polynomial on $\mathbb{F}_4$ would be invalid.

The variable p is absent from the operator coefficient $t_{S\setminus\{h\}}$ and from the first outer input h. Modulo $K = H''$, the two head variations are

$$\delta c_{pq} = t_y c_{pq} + t_p c_{qy}, \qquad \delta c_{pa} = t_y c_{pa} + t_p c_{ay}. \tag{44}$$

The two t_y terms cancel by balance. If $z = x_p$, its variation lies in H' and gives zero in the last outer slot of Θ. Changes of head lifts by K also give zero. The surviving coefficient is

$$\Theta(t_p t_{S\setminus\{h\}} c_{qy}, c_{pa}; h, z) + \Theta(t_p t_{S\setminus\{h\}} c_{pq}, c_{ay}; h, z). \tag{45}$$

Under the rank-$(d+1)$ symmetric-module identification, (45) is the image of

$$t_{[d]\setminus\{a\}}\big(c_{qy} \odot c_{pa} + c_{pq} \odot c_{ay}\big) \otimes z.$$

In the notation of the module calculation above these are respectively $C_{a,y}$ and $E_{a,y}$. Their images under Koszul multiplication are $x_p x_q$ and zero. Consequently (45) is exactly

$$\Psi_{d+1}((t_1 \cdots t_d t_y) \otimes x_p x_q \otimes z). \tag{46}$$

It vanishes by coefficient extraction.

For any off-diagonal tensor in rank $d+1$, one can choose the new generator y outside $\{p, q, z\}$ and apply this argument to the remaining d generators. Hence every off-diagonal tensor is killed. The determinant-one transvection used in Step 2 then kills every diagonal tensor. Surjectivity of (41) proves $T_{d+1} = 0$.

This is an induction on arbitrary rank, not an extrapolation from ranks five and six. Equations (44)–(46), including the zero E term, are proved in the rank-uniform module presentation.

The class bound. The previous second-central-layer theorem gives $\operatorname{cl} D_{d,2} = d+2$ for $d \geq 4$. Since $D_{d,3} \to D_{d,2}$ is central, $\operatorname{cl} D_{d,3} \leq d+3$. Any degree-$d+3$ component is contained in $M_2/M_3 = T_d$. The vanishing just proved therefore improves the upper bound to $d+2$. The previously constructed centre-by-metabelian top covers have exact generator rank d and class $d+2$, and already satisfy $H'' \subseteq Z(H)$. They give sharpness. Every at-most-d-generated algebra in the stated variety is a quotient of $D_{d,3}$, completing the third-layer case of Theorem 4.2.

E.6 The next-to-top balanced symmetric square

For $U = k^d$, write $R = \Lambda U$ and

$$P_d = \langle c_{ij} = c_{ji},\ c_{ii} = 0 \mid t_i c_{jk} + t_j c_{ik} + t_k c_{ij} = 0 \rangle_R, \qquad B_d = \operatorname{Sym}^2_R P_d.$$

Use $\deg t_i = 1$, $\deg c_{ij} = 2$. The Koszul multiplication $\mu : B_d \to R \otimes \operatorname{Sym}^2 U$ is induced by $c_{ij} \mapsto t_i x_j + t_j x_i$. For $d \geq 6$ it gives a natural identification

$$(B_d)_{d+1} \simeq \mathcal{K}_d := \ker\left(\Lambda^{d-1} U \otimes \operatorname{Sym}^2 U \xrightarrow{\partial} \det U \otimes U\right), \tag{47}$$

where

$$\partial(r \otimes xy) = (r \wedge x) \otimes y + (r \wedge y) \otimes x.$$

In particular

$$\dim \mathcal{K}_d = \frac{d(d-1)(d+2)}{2}.$$

This is a specific truncated-module computation using classical Koszul and symmetric-power constructions.

Proof of (47)**.** The bound (38) $(B_r)_n = 0$ for $n > r + 2$, together with the rank-at-most-three calculation, shows that degree $d+1$ has support of size $d-1$ or d. On a support of size $d - 1 \geq 5$ the previous top theorem identifies the component with $\det(k^{d-1}) \otimes \mathrm{Sym}^2(k^{d-1})$ and μ is injective.

For full support, the multidegree has one doubled index i. Put $O = [d]\setminus\{i\}$. Its generators are

$$X_{ab} = c_{ia}c_{ib}\, t_{[d]\setminus\{i,a,b\}}, \qquad Y_{a;bc} = c_{ia}c_{bc}\, t_{[d]\setminus\{a,b,c\}},$$

with distinct indices in O. Triangles give $Y_{c;ab} = X_{ac} + X_{bc}$ and, for four distinct indices in O,

$$X_{ac} + X_{ad} + X_{bc} + X_{bd} = 0. \tag{48}$$

The edge space on O modulo the rectangle relations (48) has dimension at most $|O|$. One way to see this is to dualize: edge weights satisfying every rectangle relation have the form $f_{ab} = u_a + u_b + c$, a space of dimension $|O|$. The images of X_{ab} are

$$Q_i + P_{i,a} + P_{i,b}, \qquad Q_i = t_{\widehat{i}} \otimes x_i^2, \quad P_{i,a} = t_{\widehat{a}} \otimes x_i x_a.$$

They have rank $|O|$: a triangle yields Q_i, and differences $P_{i,a} + P_{i,b}$ have rank $|O| - 1$. Thus μ is injective also on full support. Here $t_{\widehat{j}}$ is the product of all variables except t_j.

The kernel of ∂ has the following complete coordinate description:

- $t_{\widehat{j}} \otimes x_a x_b$ with $a, b \neq j$;
- for each doubled i, Q_i and $P_{i,a} + P_{i,b}$ with $a, b \neq i$.

Each target monomial is either one of the first items, Q_i, or $P_{i,a}$; on the latter ∂ is $(t_1 \cdots t_d) \otimes x_i$. This proves both the completeness of the displayed kernel and (47).

E.7 The depth-four source and nonlinear suspension

If $M_4 = 0$, set

$$\Theta_3(w, v; a, b, c) = [\widetilde{w}, \widetilde{v}, a, b, c].$$

As in the third-layer proof, $[M_j, H'] \subseteq M_{j+2}$ shows that the derived inputs factor through H'/H'' and the outer inputs through H/H'. Balance and the exchange of an operator with the first outer input follow from Jacobi and the partial Engel identity, now with two outer brackets appended. Peeling therefore gives, for $N \geq 8$,

$$(B_d)_{N-2} \otimes U \otimes U \twoheadrightarrow (M_3)_N.$$

At $N = d + 3$, (47) gives the natural epimorphism

$$\Phi_d : \mathcal{K}_d \otimes U^{\otimes 2} \twoheadrightarrow T_d := (M_3/M_4)_{d+3}. \tag{49}$$

Assume $T_d = 0$ and add a new generator y. Substitute $x_i \mapsto x_i + \lambda[x_i, y]$ in any old source word. On H'/H'', the coefficient operators do not change, because $[x_i, y] \in H'$. The outer inputs do not change in H/H'. Only the two commutator heads change. Consequently the resulting expression in the central target has degree at most two in λ, regardless of the number of occurrences of i in the original source word. Coefficient extraction over $\mathbb{F}_4$ is therefore valid. This detail prevents a false extraction from a raw degree-four polynomial.

The coefficient of λ induces on Koszul images the map

$$\sigma_{i,y}(z) = (t_y x_i + t_i x_y)\frac{\partial z}{\partial x_i}, \tag{50}$$

with the coefficient $r \in R$ held fixed. For example,

$$\delta_i c_{ij} = t_y c_{ij} + t_i c_{jy}, \qquad \mu(\delta_i c_{ij}) = t_j(t_y x_i + t_i x_y).$$

Triangle relations are respected: their two residual terms cancel by the triangle on (j, k, y). Thus (50) defines a map on the balanced symmetric module, not just on a list of representatives. For u, v in the old generator space,

$$\Phi_{d+1}(\sigma_{i,y}(z) \otimes u \otimes v) = 0. \tag{51}$$

E.8 The suspension images span the next source

Let $n = d + 1 \geq 7$. Let Y be the $\mathrm{GL}_n(k)$-stable span of the tensors in (51), allowing every omitted coordinate y. It lies in $\ker \Phi_n$. We show $Y = \mathcal{K}_n \otimes U^{\otimes 2}$. Fix coordinate outer inputs u, v throughout each construction.

First consider $t_{\widehat{j}} \otimes x_a x_b$, where $a, b \neq j$ and $a \neq b$. Choose $y \notin \{j, a, b, u, v\}$. In the old space use $z = t_{\widehat{j,y}} \otimes x_a x_b$ and apply $\sigma_{a,y}$. Since its coefficient already contains t_a, (50) gives exactly $t_{\widehat{j}} \otimes x_a x_b$. Thus all these tensors belong to Y. Their square counterparts follow by a shear $x_a \mapsto x_a + x_b$, choosing $a \notin \{j, b, u, v\}$ so that the coefficient and the outer inputs are fixed. Hence all $t_{\widehat{j}} \otimes x_b^2$, $b \neq j$, are in Y as well.

For $P_{i,a} + P_{i,b}$ choose $y \notin \{i, a, b, u, v\}$. In the old space the element $Q_i + P_{i,a} + P_{i,b}$ belongs to $\mathcal{K}_d$. Equation (50) with respect to x_i kills Q_i and sends the other two terms to the desired $P_{i,a} + P_{i,b}$ in rank n. Thus all these differences are in Y.

Finally fix the desired Q_j, and choose $i \notin \{j, u, v\}$. The already available missing-support tensor $t_{\widehat{j}} \otimes x_i^2 \otimes u \otimes v$ transforms under $x_i \mapsto x_i + x_j$ into the sum of itself, $t_{\widehat{i}} \otimes x_j^2 \otimes u \otimes v$, and $(Q_i + Q_j) \otimes u \otimes v$. The first two are known to lie in Y, hence so does the last. Scaling x_i by $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$ gives $(\omega^2 Q_i + \omega Q_j) \otimes u \otimes v$. The outer inputs are fixed by the choice of i. Since $\omega^2 + \omega = 1$, these two vectors separate Q_i and Q_j. This proves the spanning assertion without assuming semisimplicity. All choices are possible for $n \geq 7$.

Together with (49)–(51), this proves

$$T_d = 0 \implies T_{d+1} = 0 \qquad (d \geq 6). \tag{52}$$

E.9 The finite base and the rank-five boundary

The previous third-layer result gives $\mathrm{cl}\, D_{d,3} = d + 2$ for $d \geq 5$. Since $D_{d,4} \to D_{d,3}$ is central, $\mathrm{cl}\, D_{d,4} \leq d + 3$. In rank six the only full-support degree-nine patterns are

$$(4, 1, 1, 1, 1, 1), \quad (3, 2, 1, 1, 1, 1), \quad (2, 2, 2, 1, 1, 1).$$

Each corresponding capped graded universal extension calculation has zero top kernel. The final cochain matrices have respectively 458, 1299, and 2218 columns and full rank. The exact input tables and independent replay are supplied in `verification/central_depth/`. These are complete homogeneous cubic coefficients over $\mathbb{F}_4$, not samples at binary points.

Components supported on at most five generators vanish because $D_{5,4}$ has class at most eight. Therefore $T_6 = 0$. Induction (52) proves $T_d = 0$ for every $d \geq 6$, and centrality then improves the upper class bound from $d + 3$ to $d + 2$. The attaining models of Theorem 3.1 give equality.

For rank five, the supplied 178-dimensional capped model has

$$(\dim M_0, \ldots, \dim M_4) = (109, 60, 39, 11, 0)$$

and class eight. Its word $[[[p, q], p], [p, a], h, p, r]$ is nonzero. Its quotients by M_1, M_2, M_3 all have class seven and exact generator rank five. They give a concrete $7 \to 7 \to 7 \to 8$ chain of three central lifts. The kernels need not be one-dimensional. The same lower example and the central upper bound prove $\mathrm{cl}\, D_{5,4} = 8$ for the complete relatively free object.

Completeness of the finite starting step. At each weight, the replay enumerates all alternating cochains allowed by the multidegree cap, imposes Jacobi, every coefficient of the cubic Engel identity, and $M_4 = 0$, and computes their common kernel. It also checks that the actual added bracket coordinates are independent and span this entire kernel. Induction on weight therefore identifies the capped universal component, rather than merely verifying a model. The three degree-nine kernels vanish. The independent implementation uses the opposite pivot order and does not import the constructor. Complete cubic coefficients are required over $\mathbb{F}_4$; testing only binary-coordinate inputs would not certify the same variety.

The rank-four boundary. The supplied 40-, 41-, and 42-dimensional bracket tables form central quotients of exact rank four and classes $6, 6, 7$. Direct exact checks impose Jacobi and the complete cubic Engel coefficients. Their second derived ideals have central depths at most one, two, and three, respectively. The second-layer bound and the one-step central upper bound prove the rank-four entries of Proposition 4.3. These lower witnesses and the rank-five model establish every stated boundary value.

F Stable central depths and cubic critical layers

This appendix proves Theorems 4.4 and 4.5. Put $M_0(H) = H''$, $M_{i+1}(H) = [M_i(H), H]$, and use the balanced module B_d of Appendix E.

F.1 The common coefficient presentation

All ordinary results below hold over every characteristic-two field k with $|k| > 2$. Expand $[v, u, u, u]$ in independent scalar parameters of arbitrary Lie values. Each parameter from u has degree at most three and each parameter from v degree at most one. Evaluations over k recover every coefficient, since $|k| \geq 4$. Thus the free ordinary three-Engel algebra is the base change of the presentation over $\mathbb{F}_2$ imposing all these formal coefficients. The central-depth identities are multilinear and commute with the same base change. In particular the finite relation matrices below, whose entries are zero and one, have the same ranks over every such k. This argument does not identify the coefficient-linearized variety over $\mathbb{F}_2$ with the possibly larger ordinary three-Engel variety there.

F.2 A general peeling and propagation principle

The peeling map (39) holds for every $r \geq 1$ when $M_{r+1} = 0$ and the balanced source degree is at least six. Indeed $[M_i, H'] \subset M_{i+2}$ makes changes of either derived lift or an outer lift vanish in M_{r+1}. The partial Engel identity with inputs a, a, v exchanges a coefficient action with the first outer input. A triangle is respected by peeling from its common coefficient, which is nonempty in that degree range. This establishes the map without introducing an action of R on the target quotient.

Suppose $s \geq 5$ and

$$\operatorname{cl} D_{r,s-1} = r + 2 \qquad (r \geq a). \tag{53}$$

Then $\operatorname{cl} D_{d,s} \leq d + 3$ for $d \geq a$. For $d \geq a + 1$, put $T_d = (D_{d,s})_{d+3}$. Every weight of T_d has full support: at most $d - 1$ inputs generate an algebra of class at most $d + 2$. Peeling gives

$$(B_d)_p \otimes U^{\otimes(s-2)} \twoheadrightarrow T_d, \qquad p = d + 5 - s. \tag{54}$$

Lemma F.1 (Propagation from the actual balanced source)**.** *If $d \geq \max(a + 1, s + 2)$, then $T_{d-1} = 0$ implies $T_d = 0$.*

Proof. Omit a coordinate y and substitute $x_i \mapsto x_i + \lambda[x_i, y]$ in an old critical identity. On the two heads the first variation is

$$\delta_{i,y} c_{ij} = t_y c_{ij} + t_i c_{yj}.$$

It extends by the product rule; coefficient actions are unchanged modulo H'', and varying an outer input raises the depth by an extra unit and vanishes. Thus the span S of these source suspensions has zero image in T_d. It is $\mathrm{GL}(U)$-stable. One may extract the coefficient either from the resulting polynomial, or by projecting to the degree increased by exactly one; all higher increments are above the central class bound $d + 3$.

Fix a multilinear source on a p-element set I and a coordinate outer tail. If their combined support is deficient, its image is zero. Otherwise the tail covers all $d-p = s-5$ indices outside I and uses at most three indices of I. Choose $y \in I$ absent from the tail. A source symbol is a matching of two disjoint heads, with the other indices in its coefficient. If the matching omits y, suspending each endpoint of one head and adding the results gives the original matching, by the triangle on that head and y. If it contains y, replace that endpoint by a spare vertex; suspension relates it to a matching omitting y. Thus every multilinear source with the fixed tail lies in S. A spare vertex exists because $p \geq 7$. The old source has degree $p - 1 \geq 6$, so both peeling maps are in range.

Every degree-p presentation monomial is in the $\mathrm{GL}(U)$-span of multilinear monomials. Replace one occurrence of a repeated variable x by an unused y. The invertible shear $y \mapsto y+x$ sends the new monomial to its sum with the old one. Induction on support deficit proves the assertion; no factorial is divided out. Since $d \geq p$, enough unused variables exist. Transport an outer tensor by the inverse shear before applying this argument, so no moved tail is incorrectly held fixed. This proves that the whole source has zero image, and hence $T_d = 0$. □

This proof uses the module before Koszul multiplication. A Jacobian element can have zero first variation and still belong to the span of variations of other elements. Its individual invariance is therefore not an obstruction to this source-level propagation.

F.3 Full support, singleton symmetry, and three profiles

The weights of a full-support degree-$d + 3$ component are exactly

$$(4, 1, \dots, 1), \qquad (3, 2, 1, \dots, 1), \qquad (2, 2, 2, 1, \dots, 1).$$

If two variables occur once, shearing one into the other adds a term with deficient support, which is zero. The opposite shear also fixes the component, and three such shears give their transposition in characteristic two. All singleton permutations consequently act trivially.

Choose a singleton y. The component linear in y is the cyclic module for the associative envelope

$$A_3(H) = U(H)/\langle u^3 : u \in H \rangle$$

with the appropriate depth relations. To verify the universal statement, the semidirect bracket of an old algebra and a right module gives

$$[v + n, u + m, u + m, u + m] = nu^3 + m(u^2 v + uvu + vu^2).$$

The second coefficient is the partial polarization of the first. The free component linear in y is obtained by discarding terms with two y's, so this criterion is both necessary and sufficient there. Writing V_j for its part in M_j, one obtains

$$V_0 = (A_3(H)H)H', \qquad V_{j+1} = V_j H + A_3(H) M_j(H).$$

Both the all-factor recurrence and the equivalent ideal-generator recurrence are provided in the verification package.

To encode the other m singleton variables, use $R_m = k[t_1, \ldots, t_m]/(t_i^2)$ and replace them by $t_i Z$. An anonymous word of Z-degree $|S|$ carries coefficient t_S. The letter Z itself is not required to have cube zero. Impose cubes and repeated-value cubic coefficients only when the repeated Lie value has Z-degree zero; impose every distinct-value six-term coefficient. Repeating a positive-degree coefficient gives $t_S^2 = 0$. Distinct disjoint sets carrying the same anonymous value give even multiplicities and cancel.

This encoding preserves the complete critical component. Indeed, identifying two orders of singleton labels in a subword differs by a global singleton permutation fixing its surrounding context. If the context repeats a label the square-free product is already zero; otherwise that global permutation acts trivially as proved above. Conversely every anonymous Lie word can be labeled on any disjoint sets of the prescribed sizes. Hence the surviving cubic and depth relations are precisely those induced from the labeled component. No sum over $m!$ labelings is used. The three required old caps at rank d are therefore

$$(4, d-2), \qquad (3, 2, d-3), \qquad (2, 2, 2, d-4),$$

where the last entry counts singleton labels encoded by Z.

F.4 A cubic representation and an exterior-only alternative

The common coefficient presentation gives T_d an algebraic $\mathrm{GL}(U)$ action, with its literal integral multigrading. Full support implies that $\det(U)^{-1} \otimes T_d$ has nonnegative weights of total degree three. The classical weight criterion for polynomial representations makes it a homogeneous cubic representation [4, Section 12]. Finite-field diagonal characters alone would not establish this integral-weight statement.

If the repeated residual weight $(2, 1)$ is zero, the pure residual weight is zero too. Directly in the original $(4, 1, \ldots)$ weight, a root substitution has only a possible cubic coefficient: the linear and quadratic coefficients have the vanishing repeated shape, and the fourth has deficient support. The root-group law forces this cubic coefficient to vanish, because $(s+t)^3 + s^3 + t^3 = s^2 t + s t^2$ is not the zero polynomial. The opposite root fixes the vector, so their threefold product swaps its pure weight to a different weight and forces the vector itself to vanish.

Only the distinct cubic weights remain. Roots between two occupied indices fix each weight space, and so do roots between two unoccupied indices. Three shears show that the stabilizer permutations within either set act trivially. A root moving an occupied index to an unoccupied one has just its linear surviving coefficient. The opposite-root identity identifies this coefficient with the permutation identification between the two weight spaces. Thus

$$T_d \simeq \det(U) \otimes \Lambda^3(U) \otimes E \tag{55}$$

for a trivial multiplicity space E. This proof also shows directly, without the polynomial criterion, that permutations of the three doubled anchors act trivially whenever the repeated profile vanishes. It justifies quotienting by those permutations at the final degree; they are not imposed in lower degrees.

F.5 Exact finite starts and the stable ranges

At depth five and rank nine the three anonymous old caps are $(4, 7), (3, 2, 6), (2, 2, 2, 5)$. Complete relation elimination gives

cap	$(4, 7)$	$(3, 2, 6)$	$(2, 2, 2, 5)$
word dimension	330	4620	41580
relation rank	330	4620	41580

at old degree eleven. Hence $T_9 = 0$. Reverse-coordinate replays give the same ranks. The first two also complete the all-factor depth recurrence; the third uses the ideal-generator recurrence, with the all-factor control through the preceding degree. These are exact matrices, not sampled evaluations. Lemma F.1, with $s = 5, a = 6$, proves $\operatorname{cl} D_{d,5} \le d+2$ for every $d \ge 9$.

At depth six and rank ten the repeated cap $(3, 2, 7)$ has dimension zero. Both recurrences and reverse coordinates agree in every grade. The exterior alternative kills the pure cap and permits the top anchor-permutation quotient in $(2, 2, 2, 6)$. Its 83160 coordinates reduce to 13860 orbits. The resulting relation rank is 13860. The preceding old-degree-eleven component has dimension 31. Two complete eliminations, one using lower-multidegree equivariance and one using reversed coordinates with no such replication, agree in every grade. Thus $T_{10} = 0$. Apply Lemma F.1 with $s = 6, a = 9$. The center-by-metabelian covers attain $d + 2$ in both stable ranges. This proves Theorem 4.4.

The computations share elementary word arithmetic; they are different coordinate eliminations rather than wholly unrelated implementations. The source files, ranks, and literal lower models are included in `verification/stable_depth/`, with separate replay commands.

F.6 The two boundary representations

The complete critical weight dimensions in ranks seven and eight, ordered as pure, repeated and distinct, are respectively

$$(0, 1, 2), \qquad (1, 1, 1).$$

Their total dimensions are $7 \cdot 6 + 2\binom{7}{3} = 112$ and $8 + 8 \cdot 7 + \binom{8}{3} = 120$. For rank seven choose a nonzero repeated weight vector. Two polarizations give the independent distinct-weight vectors corresponding to $[[a, b], c]$ and $[[a, c], b]$. For rank eight each weight is a line. Singleton symmetry and the trivial one-dimensional characteristic-two character of the remaining anchor stabilizer make the transported weight bases well defined.

The explicit bracket words and all root coefficients are recorded in `check_rank7_cubic_module_actions.py` and `check_rank8_cubic_module_actions.py`. The former checks 45 root coefficients and 2160 permutation cases against the free-associative realization of $\mathrm{Lie}^3(U)$; the latter checks 54 root coefficients against $\mathrm{Sym}^3(U)$. Diagonals have the displayed multidegrees. Choose an unaffected singleton for y in each check; singleton symmetry makes the choice irrelevant. Roots, diagonals and permutations generate $\mathrm{GL}(U)$, so these intertwining maps and the equal dimensions prove Theorem 4.5. This uses the actual action, not just a coincidence of dimensions.

The usual characteristic-two sequence

$$0 \longrightarrow U^{[1]} \otimes U \longrightarrow \mathrm{Sym}^3 U \longrightarrow \Lambda^3 U \longrightarrow 0, \qquad a^{[1]} \otimes b \longmapsto a^2 b,$$

has repeated-monomial kernel of dimension d^2. It does not split for $d \ge 3$: a section on the distinct weight would send $e_1 \wedge e_2 \wedge e_3$ to $e_1 e_2 e_3$, but the shear $e_3 \mapsto e_3 + e_1$ fixes the former and changes the latter by $e_1^2 e_2$. Thus the rank-eight critical layer has a nonsplit submodule of dimension 64 and quotient of dimension 56. The cubic functors and this sequence are classical representation theory; their occurrence here identifies the two Engel boundary layers.

F.7 Literal boundary models and the complete bottom

The rank-seven and rank-eight lower models have dimensions 294 and 586 over $\mathbb{F}_4$, with operator/module decompositions $144 + 150$ and $288 + 298$. They are graded and generated by seven and eight degree-one elements. Their lower-central dimensions are

$$(294, 287, 276, 234, 164, 109, 59, 17, 3, 1, 0),$$
$$(586, 578, 565, 509, 397, 257, 187, 75, 19, 3, 1, 0),$$

and their second-derived series dimensions are $(94, 38, 10, 2, 1, 0)$ and $(187, 75, 19, 3, 1, 0)$. The tables and separately implemented verifier check Jacobi, all cubic coefficients over $\mathbb{F}_4$, generation and explicit nonzero words. They prove the lower classes 10 and 11; the central-step upper bound proves equality. Rank eight therefore excludes a stable fifth-depth start below nine.

For either lower model B, form the marked generated fiber product of B with E_d over $B/M_1(B)$. Write it as C. For $j \geq 1$,

$$M_j(C) = M_j(B) \times 0, \qquad C/M_1(C) = E_d.$$

The first assertion follows from projection onto B and the vanishing of $M_1(E_d)$; the second follows from the common quotient. Hence the quotients $C/M_j(C)$ have successive classes $d+2, d+2, d+2, d+2, d+3$. This proves the ordinary attaining chains over the complete bottom.

F.8 The sixth-depth sharpness witness

Theorem C.2 gives $J_{4,8}$ exact rank nine, dimension 597, class 12, and depth six. Hence $a_6 \geq 10$. The sixth-depth vanishing above gives the reverse inequality. The corresponding independent literal-table verifier is supplied in `verification/uniform_depth/verify_rank9_depth6.py`; its role is to check transcription of this endpoint, while the general current construction proves it over every field in the stated range.

G Optimal central realizations

Write k for a characteristic-two field with $|k| > 2$ in the ordinary part of the argument.

G.1 The center and the lower bound

Use the presentation $H_d = U \oplus P_d$ of Appendix E. Its center, for $d \geq 3$, is

$$Z(H_d) = \mathcal{S}_d \oplus \mathcal{T}_d, \qquad \mathcal{S}_d = \operatorname{span}\{t_i t_j c_{ij} : i < j\}, \qquad \mathcal{T}_d = \operatorname{span}\{T_i : 1 \leq i \leq d\}.$$

Here $T_i = t_{[d]\setminus\{j\}} c_{ij}$ is independent of $j \neq i$. The summands have dimensions $\binom{d}{2}$ and d. For a direct socle calculation, use the Koszul map $c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i$. On a multilinear support I, the triangle presentation is the edge space modulo triangle cycles, of dimension $|I| - 1$. Its image has coordinates $t_{I\setminus\{i\}} \otimes x_i$ with coefficient sum zero. Multiplication by t_i detects the ith coordinate, so no nonzero multilinear vector is central. With one doubled index i, triangles identify all partners, and the image is the line $k t_I \otimes x_i$. A missing variable acts nontrivially; only $I = [d]$ is central. With two doubled indices i, j, the symbol is $t_I c_{ij}$. A third index in I kills it by the triangle on i, j and that index. The surviving $t_i t_j c_{ij}$ is central, and projection onto H_2 detects it. These projections prove independence of all such vectors; the Koszul images prove independence of the T_i. This proves the center formula, including rank three where both summands have weight four. The same presentation gives

$$\dim H_d = (d-1)2^d + 1 + \binom{d}{2} = h_d, \qquad \operatorname{cl} H_d = d + 1.$$

Suppose $K = Z(E)$ and $E/K = H_d$. Commuting lifts define

$$\Psi : Z(H_d) \longrightarrow \operatorname{Hom}_k(H_d/H'_d, K), \qquad \Psi(z)(x) = [\widetilde{z}, \widetilde{x}].$$

Jacobi shows that a lift of z commutes with lifts of H'_d, so the map is well defined. If its value is zero, the lift is central and $z = 0$. Thus $d \dim K \geq d(d+1)/2$, proving the lower bound.

G.2 Pair detectors and the top detector

Lemma G.1 (The pair detector)**.** *The seven-dimensional algebra with basis a, b, c, u, v, w, z and table*

$$[a, b] = c, \quad [a, c] = u, \quad [b, c] = v, \quad [a, v] = [b, u] = w, \quad [a, w] = [c, u] = z$$

is ordinary three-Engel, with center and second derived ideal kz. Its quotient by kz is H_2.

Proof. Only the Jacobi triples (a, b, c) and (a, b, u) have two possibly nonzero summands; these are respectively $w + w$ and $z + z$. The other triples vanish. For the Engel identity let $X = Aa + Bb + Cc + Du + Ev + Fw + Gz$ and $Y = Pa + Qb + Rc + Su + Tv + Vw + Jz$. The c, u, v coordinates of $[Y, X]$ are

$$C_1 = PB + QA, \quad U_1 = PC + RA, \quad V_1 = QC + RB.$$

The possible w coordinate of $[Y, X, X, X]$ is $C_1AB + C_1BA = 0$. Its only other possible coordinate is at z, where it equals

$$(U_1B + V_1A)A + C_1AC = AB(PC + RA) + A^2(QC + RB) + AC(PB + QA) = 0.$$

This is a formal coefficient identity over every characteristic-two field. The center and derived assertions follow from the table. □

The scalar cocycle of this detector is $\zeta(a, w) = \zeta(c, u) = 1$, and its formal Engel obstruction vanishes.

For an ordered pair $i \to j$, pull this cocycle back along $x_i \mapsto a, x_j \mapsto b$, with all other generators zero. Call it ζ^i_{ij}. It detects exactly

$$\zeta^i_{ij}(t_it_jc_{ij}, x_i) = 1;$$

all other center–generator pairings are zero. The top vectors T_l have full support, so their two-generator projections vanish.

Let $\theta : R \to k$ extract $t_1 \cdots t_d$. The identity symmetric form gives the top cocycle

$$\begin{aligned} \alpha_I(x_i, x_j) &= 0, \\ \alpha_I(rc_{ij}, x_l) &= \theta(rt_j)\delta_{il} + \theta(rt_i)\delta_{jl}, \\ \alpha_I(rc_{ij}, sc_{kl}) &= \theta(rst_it_l)\delta_{kj} + \theta(rst_it_k)\delta_{lj} \\ &\quad + \theta(rst_jt_l)\delta_{ki} + \theta(rst_jt_k)\delta_{li}. \end{aligned}$$

It descends through the triangles and is an alternating CE cocycle, by the top-cover calculation of Theorem 3.1. That displayed construction also works at $d = 3$: completeness of the top cohomology classification in rank three is not needed here. It satisfies

$$\alpha_I(T_i, x_j) = \delta_{ij}, \qquad \alpha_I(\mathcal{S}_d, U) = 0.$$

Its ordinary Engel obstruction vanishes in rank three as well. In the notation $x = u + v, y = s + w$ with $u, s \in U$ and $v, w \in P_d$, the four obstruction terms are

$$\eta(t_uc(s, u), u) + \eta(t_ut_sv, u) + \beta(t_uc(s, u), v) + \beta(t_ut_sv, v).$$

The first has input weight four, below $d + 2$. The middle terms cancel by balance and $t_u^2 = 0$; the last is zero by the four-term formula. Thus the cocycle is three-Engel admissible for every $d \geq 3$.

G.3 Optimal packing

Put $p = \lfloor d/2 \rfloor$. Orient the complete graph so every outdegree is at most p: use the cyclic regular tournament for odd d, and delete one vertex of the $(d+1)$-vertex tournament for even d. At each vertex label its outgoing edges distinctly by $1, \dots, p$. In $K = k^p \oplus k e_{\mathrm{top}}$ use the cocycle

$$\alpha = \sum_{i \to j} e_{\mathrm{label}(i,j)} \zeta_{ij}^{i} + e_{\mathrm{top}} \alpha_I.$$

Its center-detection map is injective. The (i, label) coordinate detects one specific basis vector of $\mathcal{S}_d$; the final channel detects all of $\mathcal{T}_d$ by the identity matrix. Consequently the extension E has $Z(E) = K$ and attains the bound.

Every pair channel occurs in E'' as a bracket of c_{ij} with $t_i c_{ij}$ for an edge of that label. The top channel occurs as the bracket of c_{ij} with $t_{[d]\setminus\{j,k\}} c_{ik}$, for distinct i, j, k. Its pair pullbacks vanish. Hence $E'' = K$. Lifts of the d generators generate E: their generated algebra B has $E = B + K$, and centrality together with $K = E''$ forces $K \subset B$. The top pairing proves exact class $d + 2$.

The complete E_d maps onto this E over H_d. Its center must therefore have zero image in H_d. Since E_d'' is already central, $Z(E_d) = E_d''$ and $E_d / Z(E_d) = H_d$, proving Theorem 5.1. The graph orientation and the center-to-Hom injection are classical; the attaining cocycle gives the exact simultaneous center cost in this family.

H Local center detection for graph quotients

We prove Theorem 5.2 for the prescribed class-four quotients Q_G. The lower bound concerns every ordinary Lie realization. The table defining Q_G is metabelian: its generator actions on the derived ideal commute and have square zero, and Jacobi on three distinct generators has only zero mixed-support terms. It is therefore ordinary three-Engel. Every vertex has a neighbor, so the independent c_e detect all generator coordinates; subsequent brackets detect the c_e, u_e, v_e coordinates. Thus its center is precisely the span of the w_e, and its dimension, exact rank, and class follow from the displayed basis.

Lemma H.1 (Local restriction on every center detector)**.** *In any central extension $0 \to K \to E \to Q_G \to 0$, commuting lifts of $w_{\{i,j\}}$ and x_l give zero whenever $l \notin \{i, j\}$.*

Proof. Choose lifts X_i and put $C = [X_i, X_j]$, $U = [C, X_i]$, $W = [U, X_j]$. These project to c_e, u_e, w_e. If $l \neq i, j$, Jacobi gives

$$[W, X_l] = [[U, X_l], X_j] + [U, [X_j, X_l]].$$

The first inner bracket is in K since the corresponding mixed action in Q_G is zero. Write $D = [X_j, X_l]$. The second term is

$$[[C, X_i], D] = [[C, D], X_i] + [C, [X_i, D]].$$

Both inner brackets on the right are in K: Q_G is metabelian and $[x_i, [x_j, x_l]] = 0$ on three distinct labels. All terms therefore vanish. Different lifts differ by elements of the central kernel, so the conclusion is independent of the chosen expressions. □

Proof of the lower bound in Theorem 5.2. Now take $K = Z(E)$. The classical injection is

$$\Psi : Z(Q_G) \hookrightarrow \mathrm{Hom}(Q_G / Q_G', K), \qquad \Psi(z)(\bar{x}) = [\widetilde{z}, \widetilde{x}].$$

Jacobi makes the pairing zero on Q'_G, and injectivity follows because a lift commuting with every lift is central. For $W \subseteq V$, Lemma H.1 says that the entire $|\mathcal{E}(G[W])|$-dimensional span of its edge centers has detectors supported only on the $|W|$ generator coordinates in W. Thus

$$|\mathcal{E}(G[W])| \le |W| \dim K.$$

Taking all nonempty W gives the claimed stronger local lower bound. It applies to all Lie realizations, without imposing Engel on E. □

Classical orientation input, with its proof. A graph has an assignment of each edge to one of its endpoints with at most p edges assigned to any vertex if and only if $|\mathcal{E}(G[W])| \le p|W|$ for every W. To see sufficiency, form a bipartite incidence graph whose left vertices are edges and whose right vertices are p copies of each vertex. For a set F of edges its endpoints form a set W and $|F| \le |\mathcal{E}(G[W])| \le p|W|$; Hall's criterion gives a matching covering all edges. Necessity is immediate. This is the bounded-outdegree orientation criterion [11]; the matching proof makes the input explicit.

Attainment in Theorem 5.2. Take $p = p(G)$, choose the preceding endpoint assignment, and label the edges assigned to each vertex injectively by $1, \dots, p$. Let $K = k^p$ with basis $z_1, \dots, z_p$. For an edge $e = \{i, j\}$ assigned to i, map Q_G onto H_2 by $x_i \mapsto a$, $x_j \mapsto b$ and all other generators to zero. Pull back the scalar cocycle of Lemma G.1 and put it in the channel $z_{\ell(e)}$. Sum these cocycles over the edges. The result is an alternating CE cocycle, and its Engel obstruction vanishes: the obstruction $\alpha([Y, X, X], X)$ is linear in the central cocycle, and vanishes for each pair pullback by the lemma.

The detector of w_e has exactly one nonzero coordinate, $(i, \ell(e))$, and different edges have different such coordinates. Hence the center-detection map is injective, so the resulting extension has center exactly K and quotient by its entire center equal to Q_G. Every channel can be taken to occur: if the chosen orientation used fewer than p labels at every vertex it would contradict the minimal value of p. At a vertex of load p use all labels there. For each channel some edge assigned to an endpoint i then gives $[c_e, [c_e, x_i]] = z_{\ell(e)}$. Thus $E'' = K$. Lifts of the d generators generate E, since their generated subalgebra B has $E = B + K$ and $K = E'' \subseteq B$. Finally, $[w_e, x_i] = z_{\ell(e)} \neq 0$ gives class at least five; centrality of K and class four of Q_G give equality. □

Corollary H.2. *For the complete graph on $d \ge 2$ vertices the minimum is $\lceil (d-1)/2 \rceil$. For any nonempty forest without isolated vertices it is one. The graph $K_4 \sqcup K_2 \sqcup K_2$ has minimum two, although the single global center-to-Hom dimension estimate gives only $\lceil 8/8 \rceil = 1$.*

I Proof inputs and reproducibility

The uniform upper bound in Theorem 4.1 has a general Frobenius-jet proof. A transcription control checks its auxiliary anchor-degree-four, singleton-degree-three component at depth two in 35 associative coordinates: it vanishes, while removal of the partial identity leaves dimension three. This is not a proof input.

The current controls check generation, all-factor M_u filtrations, centers and the end-of-block truncation. The affine check verifies 44253 basis pairs in dimension 298; the cubic checks verify the four-term root coefficients; the graph controls check 49 constructions. Their general statements have separate symbolic proofs.

The exact low-depth inputs are retained: the three rank-six systems, the fifth-depth start, and the sixth-depth exterior system with 13860 final coordinates after the proved anchor quotient. Appendix F gives their relation ranks. Literal models and generator substitutions check endpoint classes and representations.

All supporting files are in `verification/`. Run

```
python verification/run_replay.py --output-dir replay_new
```

and add `--extended` for the expensive unchanged endpoint systems. The runner uses an isolated copy and records each actual invocation. New controls and endpoint materials are in `verification/uniform_depth/`. A proof-to-certificate map, source fingerprints, and fresh versus inherited replay scopes are included in the independent package. All controls use the Python standard library.

References

[1] L. G. Kovács and R. Stöhr, Free centre-by-metabelian Lie algebras in characteristic 2, *Bulletin of the London Mathematical Society* **46** (2014), no. 3, 491–502. doi:10.1112/blms/bdu011. Author manuscript: `https://archives.maths.anu.edu.au/people/Kovacs/K114.pdf`.

[2] Yu. V. Kuz'min, Free center-by-metabelian groups, Lie algebras, and D-groups, *Math. USSR-Izv.* **11** (1977), no. 1, 1–30. doi:10.1070/IM1977v011n01ABEH001443.

[3] N. Mansuroğlu and R. Stöhr, Free centre-by-metabelian Lie rings, *The Quarterly Journal of Mathematics* **65** (2014), 555–579. doi:10.1093/qmath/hat017.

[4] O. Mathieu, Tilting modules and their applications, in *Analysis on Homogeneous Spaces and Representation Theory of Lie Groups*, Advanced Studies in Pure Mathematics **26** (2000), 145–212. Section 12. `https://www.imsc.res.in/~knr/vdk/mathieu.pdf`.

[5] Ş. Papadima and A. I. Suciu, Chen Lie algebras, *International Mathematics Research Notices* **2004** (2004), no. 21, 1057–1086. doi:10.1155/S1073792804132017.

[6] H. Strade, *Lie algebras of small dimension*, preprint, 2006. `https://arxiv.org/abs/math/0601413`.

[7] G. Traustason, Engel Lie-algebras, *The Quarterly Journal of Mathematics* **44** (1993), no. 3, 355–384. doi:10.1093/qmath/44.3.355.

[8] G. Traustason, A polynomial upper bound for the nilpotency classes of Engel-3 Lie algebras over a field of characteristic 2, *Journal of the London Mathematical Society* **51** (1995), no. 3, 453–460. doi:10.1112/jlms/51.3.453.

[9] R. Stöhr, *Bases, filtrations and module decompositions of free Lie algebras*, author preprint, MIMS EPrint, 2007, Section 2. `https://eprints.maths.manchester.ac.uk/703/1/Decom_submit.pdf`.

[10] B. Lancellotti and Th. Weigel, *The projective indecomposable modules for the restricted Zassenhaus algebras in characteristic 2*, arXiv:1409.4310v3, 2015. `https://arxiv.org/abs/1409.4310`.

[11] S. L. Hakimi, On the degrees of the vertices of a directed graph, *Journal of the Franklin Institute* **279** (1965), 290–308. doi:10.1016/0016-0032(65)90340-6.

[12] B. Li, *Quadratic orbit maps and three-Engel Lie algebras in characteristic two*, companion manuscript, dated revision, 2 November 2026. Earlier public version: doi:10.5281/zenodo.23874202.

[13] B. Li, *Extremal dimensions and critical isomorphism growth of metabelian three-Engel Lie algebras*, companion manuscript, dated revision, 2 November 2026. Earlier public version: doi:10.5281/zenodo.23874228.

[14] B. Li, *Quadratic maps satisfying $f \circ f = f + \mathrm{id}$: nilpotence, Lie reconstruction, and growth*, unified research manuscript, 2026. Historical public version: doi:10.5281/zenodo.23852642. Frozen 209-page version, 2 November 2026: doi:10.5281/zenodo.23874173.